

Estimating the burial depth of conventional explosives

A hindcast of sandwaves on the Dutch Continental Shelf



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Client	Rijkswaterstaat – TenneT TSO BV
Contact	Henk Neggers – Robert Koens
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Author(s)

	Tom Roetert	
	Tommer Vermaas	
	Ruben White	

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Summary

Rijkswaterstaat and TenneT have requested Deltares to perform a seabed mobility analysis for the Dutch Continental Shelf (DCS). This study modifies and extends the analysis performed in the Deltares (2020a) pilot study for the DCS by applying a more detailed methodology and additional bathymetrical data.

The overall objective of the Deltares activities is to generate one or more data products (i.e., maps and GIS files) that will indicate at what depth below the seabed objects can be expected and what the expected future seabed levels will be. This information will simplify or preclude the choice of means for mitigation measures for Conventional Explosives (CE) and offshore activities in the subsurface – such as cable trenching or sand mining – depending on the location and situation.

The aim of this morphodynamic assessment is to characterise the seabed dynamics and to quantify seabed level changes over the period 1945 to 2022 and over the period 2022 to 2050. Different levels will be provided describing the bandwidth in Lowest Seabed Levels for given years. This study focusses solely on the areas where sand waves are present on the DCS. In areas offshore without sand waves the sea floor dynamics are very low and not predictable on this long timescale.

The DCS is characterised by significant areas of sand waves particularly in the southern part of the DCS. Deepest water is found along the southwestern edge of the sand wave region. As expected, the water depth gradually decreases towards the coast. The area is furthermore characterised by tidal sand banks in the north and south, with the northern banks running from north to south and the southern banks facing the coast of Zeeland. In the southwestern part of the sand wave area, long bed waves with wavelengths of around 1.5 km can be seen. These rhythmic features are larger than sand waves, which typically have lengths ranging from 100 to 1000 m.

To analyse seabed dynamics a detailed analysis is presented focussing on sand waves present in the DCS. Sand wave dynamics are determined by using a 2D cross-correlation technique. The sand waves typically migrate to the north-east, and locally to the south-west, with rates mostly ranging from 0 to 3 m/year. In the south-western part of the sand wave area, migration rates are low. Meanwhile, the north-eastern region shows higher migration rates which gradually increase closer to the Wadden Islands. The highest migration rates are found near coastal areas and on top of sand banks, which significantly redirect the migration direction of the sand waves in several areas.

Based on the extrapolation of seabed trends, several seabed levels have been derived for the period 1945 to 2022 and for the period 2022 to 2050. These levels describe the hindcasted and predicted bandwidths in the Lowest Seabed Levels by means of a lower, mean and upper prediction. Furthermore Best-Estimate Bathymetries are provided for the years 2022 and 2050.

The analysis provides insight in the dynamics of sand waves on the DCS. The quality of the results is subject to a number of limitations. For example, limited spatial coverage of survey data, quality of available data and human interventions can significantly influence results. Quality checks resulted in exclusion of certain datasets because of limited quality.

The predicted seabed level changes presented in this study follow from the applied morphological analysis techniques, describing the (uncertainty of the) physics and the natural variability of the analysed morphological system. No additional safety margins for design purposes have been applied.

The presented hindcasted and predicted future seabed levels are a result of the applied methodology. This methodology was previously successfully applied in detailed morphodynamic assessments for wind farm zones on the DCS. Because of local characteristics, upscaling of the analysis to the entire DCS resulted in increased uncertainties of historic and future seabed levels as compared to detailed, local morphodynamic assessments. The presented seabed levels provide a good insight in historic and future bandwidths of seabed levels, however detailed assessments are recommended in case depths of buried objects and cable burial depths need to be determined with more accuracy.

The presented seabed levels are a valuable dataset to (further) decrease the risks related to CE. Despite the limitations, as can be expected from a hindcast of 60 to 75 years, the resulting surfaces and maps will help in assessing the likelihood of presence of CE at and under the seafloor.

Samenvatting (Nederlands talig)

Rijkswaterstaat en TenneT hebben Deltares gevraagd een morfodynamiek studie uit te voeren voor het Nederlands Continentaal Plat (NCP). Deze studie vormt een uitbreiding op de analyse die is uitgevoerd in de Deltares pilotstudie (2020a) voor het NCP, door een meer gedetailleerde methodologie toe te passen en aanvullende bathymetrische gegevens te gebruiken.

Het algemene doel van deze studie is het genereren van een of meer dataproducten (kaarten en GIS-bestanden) die aangeven op welke diepte onder de zeebodem objecten kunnen worden verwacht en wat de verwachte toekomstige zeebodemniveaus zullen zijn. Deze informatie zal de keuze van mitigerende maatregelen voor Conventionele Explosieven (CE) en offshore activiteiten in de ondergrond – zoals het ingraven kabels of zandwinning – vereenvoudigen of uitsluiten, afhankelijk van de locatie en situatie.

Het doel van deze morfodynamiek studie is om de dynamiek van de zeebodem te karakteriseren en veranderingen in de zeebodem te kwantificeren over de periode 1945 tot 2022 en over de periode 2022 tot 2050. Er zullen verschillende niveaus worden gegeven die de bandbreedte in laagste zeebodemniveaus voor bepaalde jaren beschrijven. Deze studie richt zich uitsluitend op de gebieden waar zandgolven aanwezig zijn op het NCP. In gebieden buiten de kust zonder zandgolven is de dynamiek van de zeebodem erg laag en niet voorspelbaar op deze lange tijdschaal.

Het NCP wordt gekenmerkt door aanzienlijke gebieden met zandgolven, met name in het zuidelijke deel. De grootste waterdiepte wordt aangetroffen aan de zuidwestelijke rand van het zandgolfgebied, de waterdiepte neemt geleidelijk af richting de kust. Het gebied wordt verder gekenmerkt door getijdenzandbanken in het noorden en zuiden, waarbij de noordelijke zandbanken van noord naar zuid lopen en de zuidelijke oevers gericht zijn op de Zeeuwse kust. In het zuidwestelijke deel van het zandgolfgebied zijn *long bed waves* bodemvormen te zien met golflengten van ongeveer 1,5 km. Deze bodemvormen zijn groter dan zandgolven, die typisch een lengte hebben van 100 tot 1000 m.

Om de dynamiek van de zeebodem te analyseren wordt een gedetailleerde analyse gepresenteerd waarbij de nadruk ligt op zandgolven die aanwezig zijn in het NCP. Zandgolfdynamiek wordt bepaald door gebruik te maken van een 2D kruiscorrelatietechniek. De zandgolven migreren typisch naar het noordoosten en plaatselijk naar het zuidwesten, met snelheden die meestal variëren van 0 tot 3 m/jaar. In het zuidwestelijke deel van het zandgolfgebied is de migratiesnelheid laag, terwijl de noordoostelijke regio hogere migratiesnelheden laat zien die geleidelijk toenemen richting de Waddeneilanden. De hoogste migratiesnelheden worden gevonden in de buurt van kustgebieden en op zandbanken, waardoor de migratierichting van de zandgolven in verschillende gebieden aanzienlijk wordt beïnvloed.

Op basis van extrapolatie van zeebodemtrends zijn voor de periode 1945 tot en met 2022 en voor de periode 2022 tot en met 2050 verschillende bodemniveaus afgeleid. Deze bodemniveaus geven een bandbreedte in de *hindcast* en de extrapolatie naar de toekomst van het Laagst Opgetreden Bodemniveau, door een lage, gemiddelde en hoge voorspelling te geven. Daarnaast worden de bodemniveaus van de bathymetrieën voor de jaren 2022 tot en met 2050 bepaald.

De analyse geeft inzicht in de dynamiek van zandgolven op het NCP. De kwaliteit van de resultaten heeft echter een aantal beperkingen. Zo kunnen beperkte ruimtelijke dekking van meetdata, de kwaliteit van de beschikbare gegevens en menselijke ingrepen de resultaten aanzienlijk beïnvloeden. Kwaliteitscontroles resulteerden in uitsluiting van bepaalde datasets vanwege beperkte kwaliteit.

De veranderingen van de zeebodem die in deze studie zijn bepaald volgen uit de toegepaste morfologische analysetechnieken, die de (onzekerheid van de) fysica en de natuurlijke variabiliteit van het geanalyseerde morfologische systeem beschrijven. Er zijn geen aanvullende veiligheidsmarges voor ontwerpdoeleinden toegepast.

De gepresenteerde *hindcast* en voorspelde toekomstige bodemniveaus zijn het resultaat van de toegepaste methodes. Deze methodiek is eerder met succes toegepast in gedetailleerde morfodynamische analyses voor windparkgebieden op het NCP. Vanwege lokale kenmerken resulteerde opschaling van de analyse naar het gehele NCP in grotere onzekerheden van historische en toekomstige bodemniveaus in vergelijking met gedetailleerde, lokale morfodynamische analyses. De gepresenteerde bodemniveaus geven een goed inzicht in historische en toekomstige bandbreedtes van zeebodemniveaus, maar gedetailleerde beoordelingen worden aanbevolen voor het geval de diepte van begraven objecten en kabels nauwkeuriger moeten worden bepaald.

De gepresenteerde bodemniveaus zijn een waardevolle dataset om de risico's gerelateerd aan CE (verder) te verminderen. Ondanks de beperkingen, zoals kan worden verwacht op basis van een *hindcast* van 60 tot 75 jaar, zullen de resulterende oppervlakken en kaarten helpen bij het beoordelen van de waarschijnlijkheid van aanwezigheid van CE op en onder de zeebodem.

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1 Introduction

1.1 Project background

Safety in relation to conventional explosives (CE) is an important issue during preparatory work of seabed disturbing activities and working on infrastructure on the seabed of the North Sea. These CE were dropped on the seabed during the First (WWI) or Second World War (WWII) or in the period since WWII and may have been buried by the changing seabed or may still be present close to the surface. Knowledge of the position of potentially present explosives in the subsurface is important for risk management.

In the context of this issue, Deltares performed a study for Rijkswaterstaat in 2019 to determine the base of the active layer for all areas on the Dutch Continental Shelf (DCS) where sand waves occur (Deltares, 2020a). Figure 1.1 gives an overview of the areas of the DCS where sand waves occur. Note that the active layer only describes the area under the sand waves and not the base of the Holocene layer. Determining the base of the active layer can be seen as an approximation of the locations of the sand wave troughs over a period where the sand wave has migrated at least a full wavelength (T3 in Figure 1.2). This 2020 study was limited in scope and resulted in an initial result that, while providing useful insights, is not directly applicable for mitigation measures for CE. To arrive at a more detailed product, Rijkswaterstaat and TenneT have asked Deltares to modify and extend the analysis. In addition it was requested to extend the period of interest towards the future to highlight impacts on future wind farm development.

To highlight the relevance of sand waves in relation to infrastructure Figure 1.1 gives an overview of the areas of the DCS where sand waves occur together with locations of current and future wind farms, their export cables and sand mining areas. It is noted that the figure does not show other relevant infrastructure such as pipelines, telecom cables and oil and gas platforms. From the figure it can be seen that a significant amount of infrastructure is located within areas of sand waves.

The burial depth of a CE at a specific location in the North Sea depends on the morphology (i.e., the presence of mega ripples, sand waves and sand banks) and the morphodynamics (i.e., the speed at which the present seabed features migrate). In areas where there are no sand waves present, the CE is expected to remain at or close to the surface. Here, only smaller bedforms are present, usually a few decimetres in height.

In areas with sand waves, the CE might become buried due to the migration of these larger bedforms. In the long term the CE is expected to descend to the depth at which the passing troughs of the sand waves are located. This depth is also referred to as the 'base of the active layer'. An important assumption here is that the large-scale underlying bathymetry remains stable over the period of several decades, and that there are no erosion-resistant (clay/peat) layers at this depth. If these erosion-resistant layers are present, then the top of these layers are the base of the active layer. However, this cannot be determined based on morphological analyses, but could be determined based on seismic and borehole data. The latter is not part of the study described in this report. Since sand waves in principle occur in areas with sand, the presence of erosion-resistant layers within a sand waves is expected not to occur often.

The sand waves in the North Sea have different dimensions, depending on their location. Moreover, the speed at which these sand waves travel and the direction in which they travel also vary spatially. For this reason, the CE from World War I and World War II may not have

been buried to the base of the active layer at all locations. Figure 1.2 contains a schematic representation of how CE are buried as sand waves migrate over time. T1 and T2 illustrate situation in which CE may still be present at a shallower depth below the seabed. Note that the previous study by Deltares (2020a) assumed that all CE were located on the base of the active layer (T3 in Figure 1.2).

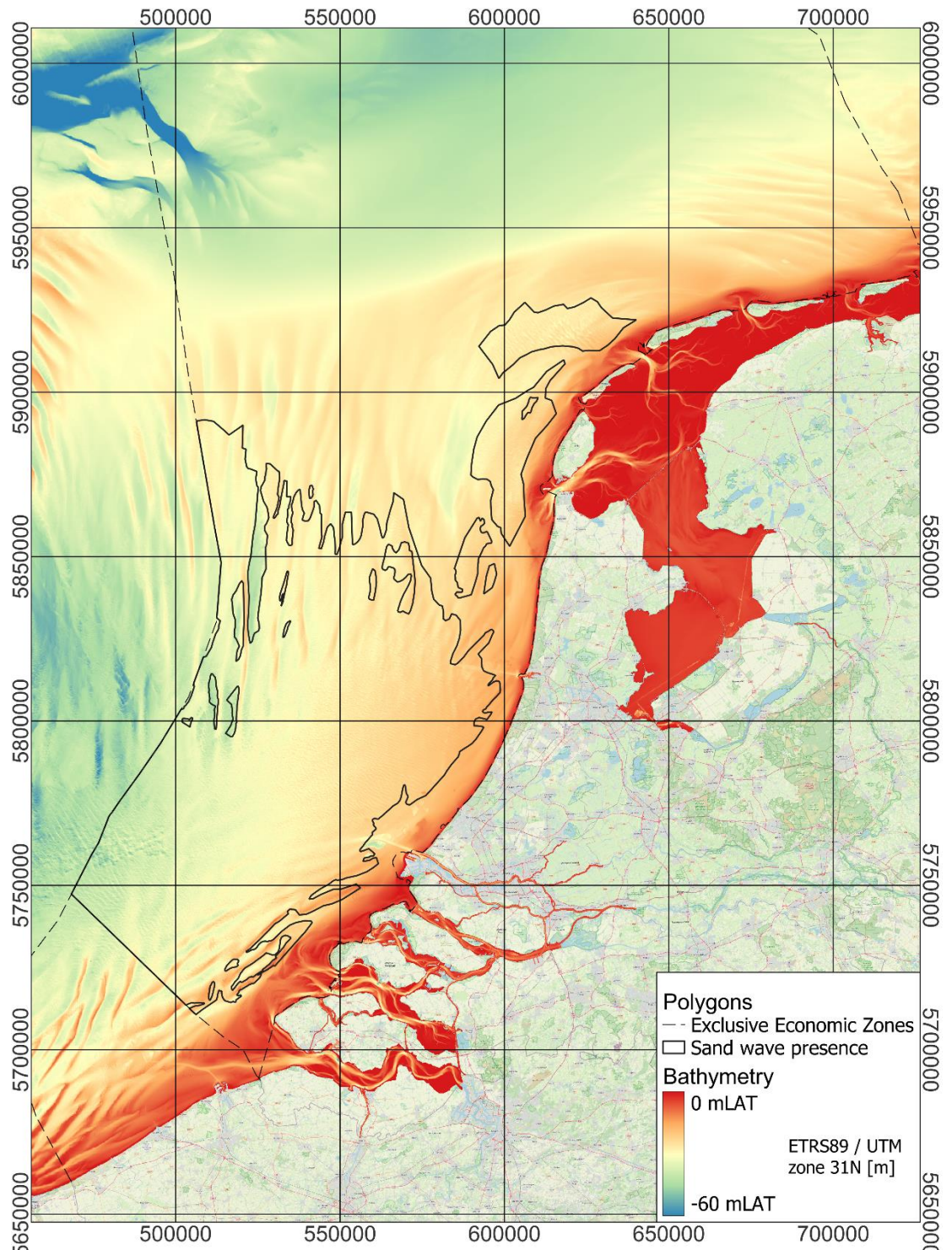


Figure 1.1: Overview of the areas of the DCS where sand waves are present including relevant infrastructure such as wind farm zones (green and blue polygons), export cables (blue lines) and sand mining areas.

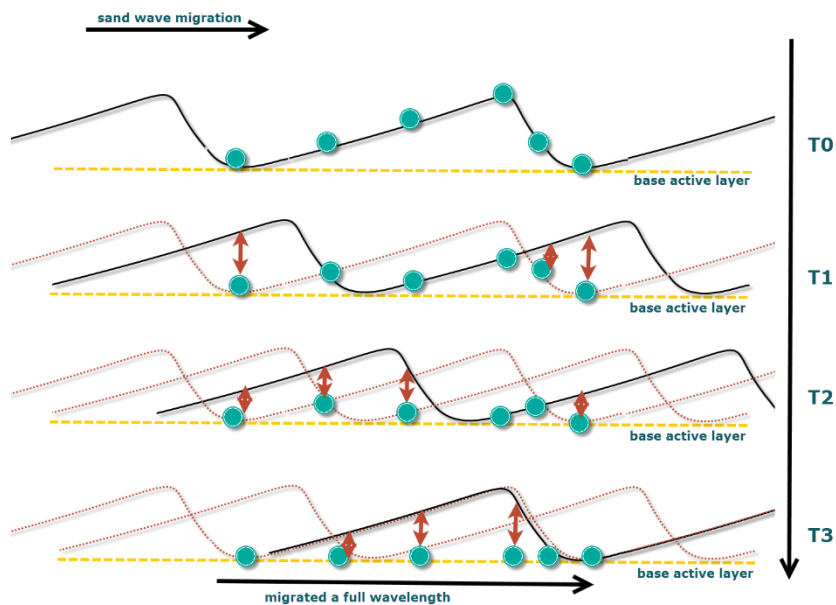


Figure 1.2: Schematic representation of sand wave migration and burial of CE to the base of the active layer (cross section of bathymetry).

1.2 Objectives and deliverables

Rijkswaterstaat and TenneT have requested Deltares to perform a seabed mobility analysis for the DCS. This study modifies and extends the analyse performed in the Deltares (2020a) study for the DCS by applying a more detailed methodology and additional bathymetrical data.

The overall objective of the Deltares activities is to generate data products (i.e., maps and GIS files) that will simplify or preclude the choice of means for mitigation measures for CE, depending on the location and situation. Moreover, the data products will also indicate at what depth below the seabed objects can be expected. This objective is to be achieved by answering the following sub-questions:

- Where are CE potentially present on the seabed surface?
- Where is the thickness of the sediment above an expected CE a minimum of 2.5, 4, 5, 6, 8 and 10 metres?
- Where is the thickness of the sediment above an expected CE a maximum of 5 metres?
- What are the seabed levels in 2022 and 2025-2050 (every 5 years)?

The seabed mobility analysis is performed for CE that ended up on the seabed of the DCS in the period 1914-1918 (WWI) and 1940-1945 (WWII). CE moved or dropped on the seabed after WWII are thus not considered. The following products are delivered:

- An overview of the data that have been used;
- The following data files for 1945, with all being estimates of the lowest hindcasted seabed level between 1945 and the most recent measurement:
 - Lowest Object Level (LOL);
 - Best-Estimate Object Level (BEOL);
 - Highest Object Level (HOL).
- The following data files for 2022 and from 2025 to 2050 (every 5 years)¹:
 - Low-Estimate of the lowest SeaBed Level (LSBL);
 - Best-Estimate of the lowest SeaBed Level (BSBL);
 - High-Estimate of the lowest SeaBed Level (HSBL).
- Map and ArcGIS file (raster format) with qualification of results, for example with the following classes: not enough bathymetry data, rejected during quality control, etc.;
- Map and ArcGIS file with 'zones' of expected CE depth:
 - CE expected on surface of seabed;
 - CE not expected in the upper 2.5, 4, 5, 6, 8, and 10 metres below seabed.
- A report describing the methodology, results and data files (this report).

1.3 Reader

This report is structured as follows. Chapter 2 gives an overview of the available data. The methodology for the seabed mobility analysis is described in Chapter 3. The main results of this project are the datasets, of which maps are presented in Chapter 4. Finally, recommendations are given in Chapter 5.

¹ All in ArcGIS file (raster) format, with the coordinate system UTM31N ETRS89 and vertical reference plane LAT, with a resolution of 25x25 metres.

2 Overview of available data

A complete list of used bathymetric surveys is delivered in a separate overview. In this chapter a short summary of the data is given, for each source.

2.1 Netherland Hydrographic Office (NLHO)

From the NLHO in total 4080 surveys were available for this study, measured between 1967 and 2021. Of all surveys 3559 were selected for this study, other surveys had too low resolution or did not cover the area where sand waves occur. An overview of the number of surveys per location for the entire DCS from the NLHO is given in Figure 2.1.

The data are recorded using single beam and multi beam echosounding devices and are delivered to Deltares at 3x5 m resolution. They were interpolated to 5x5 m grids the using inverse distance weight method, in blocks of 5x5 km to keep the file size small enough for the interpolation since some surveys cover a very large area. All metadata can be found at <https://opendap.deltares.nl/thredds/catalog/opendap/hydrografie/surveys/catalog.html>. Original data requested through <https://www.defensie.nl/onderwerpen/hydrografie/>.

The following surveys have been excluded from the final calculations due to deviating results: sid386, sid1093, sid10425, sid10426, sid10427, sid10428, sid8764, sid1921, sid13598 and sid15052. Furthermore, provided data for the surveys conducted in 2021 and 2022 were of low quality. These are only included in determining trends in seabed dynamics but not considered during extrapolation of the data.

2.2 Rijkswaterstaat

Rijkswaterstaat CIV delivered 125 surveys performed in sand mining areas to use in this project. Out of these, 52 surveys were suitable to use for the calculations. The other surveys were excluded because of:

- Absence of sand waves;
- Dredging activities visible in the seabed morphology of the survey;
- Duplicates in the delivered data;
- Bad data quality or too small area.

If needed, surveys were converted from NAP to LAT vertical reference datum using the reduction matrix (https://english.defensie.nl/downloads/applications/2020/06/12/nllat2018). The data were delivered as grids with a resolution of 5x5 or 1x1 m.

2.3 TenneT

From TenneT multi beam grids covering two export cable trajectories have been received and used in this study. The data has a resolution of 0.30x0.30 m and 2x2 m.

2.4 RVO (Netherlands Enterprise Agency)

For the windfarm zones that are investigated by RVO, multi beam datasets are available at <https://offshorewind.rvo.nl/>. The datasets were acquired through this site for the following windfarm zones:

- Borssele
- Hollandse Kust (zuid);
- Hollandse Kust (noord);
- Hollandse Kust (west);
- IJmuiden Ver.

The data were downloaded as grids with a resolution of 0.50x0.50 m or 1x1 m.

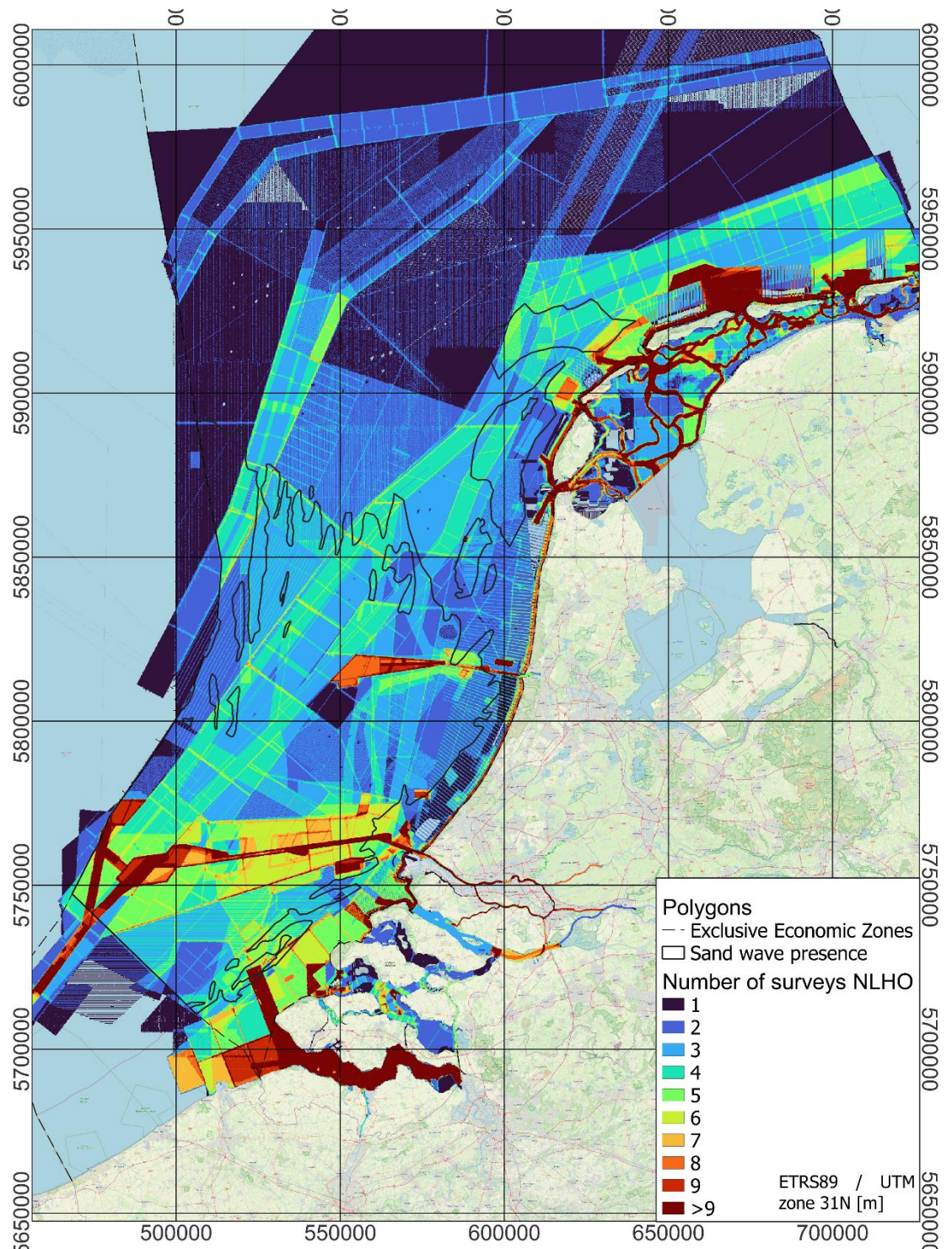


Figure 2.1: Number of surveys conducted per location based on NLHO data.

3 Methodology

In this chapter, the methodology of the seabed mobility analysis for the DCS is presented. This analysis comprises of seven main steps:

- Processing of available data (Section 3.1);
- Splitting of bedforms (Section 3.2);
- Analysis of sand wave dynamics (Section 3.3);
- Extrapolation of historic trends (Section 3.4) and
- determination of uncertainties (Section 3.5) to
- Calculate the future and historic seabed levels including uncertainties (Section 3.6);
- Translation of historic seabed levels to expected coverage of objects (Section 3.7).

Detailed descriptions of each of these steps are given in Sections 3.1 to 3.6, respectively. In Section 3.8 the main quality checks that were performed are described. Sections 3.1 to 3.6 are based on Deltares (2023) where these methods were used as well. The methods have been adjusted for this study to be applicable for the entire DCS but their basic working as described in this chapter remains the same as in the referred study.

3.1 Processing of available data

The NLHO data was already available at Deltares before the start of this project. Those datasets were processed into 5x5 km blocks to be able to handle the sometimes very large datasets. The data were interpolated to 5x5 m using inverse distance weight method. For this study the latest surveys from the NLHO were added to the dataset.

For this study the division in 5x5 km blocks was also useful, to be able to process the entire DCS. Therefore, all other surveys were processed to match the NLHO data, as follows:

- Make all values uniform, e.g. being negative numbers relative to LAT and in ascii grid format;
- Resample the data to 5x5 m (all had smaller resolution) and the same grid alignment and coordinate system;
- Split all the grids to the 5x5 km blocks;
- Create polygons of the coverage of each survey and add the metadata (start and end date of the survey).

3.2 Morphodynamic overview and splitting of bedforms

In the DCS three main bedform types are present, from large to small scale these are: sand banks, sand waves and megaripples. The sand banks are large ridges with wavelengths in the order of 1-10 km and are relatively stable bedforms with low migration rates. In many cases on the DCS the sand banks are covered with sand waves. The sand waves are dynamic bedforms that move with 0-20 m/year and have lengths in the order of 100-1000 m and heights in the order of a couple of metres (Van Dijk, 2011). The smallest scale bedforms considered in this study, the megaripples, can be superimposed on sand waves and sand banks. They have lengths in the order of metres and heights up to approximately 1 m.

The different types of bedforms were separated to analyse them individually. Separate layers are created for the sand waves, the large-scale underlying seabed and megaripples. First, the migrating part of the bathymetry has been separated from the underlying, large-scale bathymetry – the sand banks and overall trends in the seabed. For this purpose, a 2D high pass filter was applied on the available surveys. The filter size was chosen such that the mobile bedforms (i.e. sand waves and megaripples) could be separated, while the underlying bathymetry remains unaltered in shape and is not noticeably smoothed by the filtering process. A similar process is repeated to split the sand waves and the megaripples.

The specific settings of the filter are obtained by a Fourier approximation of the wavelengths. Results are analysed by comparing gradients present in the underlying large-scale bathymetry indicating still present sand wave information. Furthermore, the mobile bathymetries are checked by the extent to which the large-scale seabed was present. A perfect mobile bathymetry should depict only bedforms fluctuating around zero. The filtering for which splitting of bedforms provided the best results is used to derive the layers with the various bedforms.

To prevent effects at the edges of surveys the outer 100 m of surveys is used as a buffer. Similarly, around the 5x5 km processing blocks, a 2,5 km buffer is applied. This is a much larger distance to also form a suitable buffer for the analyses of the migration rates (Section 3.3).

After processing of the individual 5x5 km blocks the resulting layers have been merged in and areas with different morphological settings have been visually inspected. On the individual blocks 2D Fourier power plots have been created, to analyse if the separating of the bedforms was performed well with the chosen filtering.

The resulting layers (Table 3.1) form the basis for the data-driven analysis and for assessing future and historical seabed levels, as discussed in Section 3.4.

Table 3.1: Definitions of various bathymetrical data fields used in this study.

Short name	Description	Large-scale seabed	Sand waves	Megaripples
Bathymetry	Full measured bathymetry including all bedforms	✓	✓	✓
Large-scale bathymetry	Filtered bathymetry with the large-scale seabed only	✓	✗	✗
Mobile bathymetry	Filtered bathymetry with sand waves and megaripples only	✗	✓	✓
Sand wave field	Filtered bathymetry with sand waves only	✗	✓	✗
Megaripple Field	Filtered bathymetry with megaripples only	✗	✗	✓

3.3 Analysis of sand wave dynamics

To derive historical trends of seabed dynamics, a data-driven analysis is performed for each of the bedform types. This section focuses on the analysis of sand waves which are, considering their timescales of migration (order of years) and their dimensions (several metres high), the most relevant for the reconstruction of the 1945 seabed. In this, the following three substeps are included:

- Determination of sand wave analysis locations;
- Determination of sand wave dynamics;

- Determination of sand wave dimensions.

In the analysis, the sand wave field is used for the first two steps as this excludes noise in the signal because of the presence of the smaller-scale bedforms. For the third step, mobile bathymetry describes the whole sand wave heights. Each of the substeps is further elaborated in more detail in the remainder of this section.

3.3.1 Sand wave analysis locations

In the sand wave analysis, an approach explicitly focussing on quantifying the dynamics and characteristics of the individual sand waves is applied. First, sand wave crests are identified, and several analysis locations are defined along these crests, spaced approximately 50 to 100 metres apart. Similarly, the detection of crests is performed by using a 1D Fourier assessment similar as was done in the morphodynamic assessments for Hollandse Kust (west) and IJmuiden Ver (Deltares, 2020b, 2023). A grid for the analysis area from the available bathymetry data is created with the columns roughly perpendicular to the sand waves. From this grid each column is analysed, and local maxima (crests) and minima (troughs) were extracted. Each sand wave is analysed at several locations indicating spatial variability along each sand wave. The analysis locations defined are used for further sand wave analysis.

3.3.2 Sand wave dynamics

Sand waves are generally characterised by a mild sloping stoss side and a steeper lee side oriented in the direction of propagation. This distinction may be less pronounced in areas where sand waves can migrate in opposing directions. Furthermore, sand waves tend to migrate in a direction roughly perpendicular to the crest with a constant form in the direction of the steepest gradient of the sand wave stoss side or in the direction of the residual current. This may be less pronounced in areas where the underlying large-scale seabed shows a highly non-uniform morphology. This knowledge is used to analyse the sand wave dynamics and provides a good first indication of the dynamics of the seabed as a result of sand wave migration.

To analyse sand wave dynamics, the mobile bathymetries of all available surveys are analysed by comparing all bathymetries to each other. All analysis locations are analysed for every possible comparison, indicating both spatial and temporal (only when the number of surveys at a given location is larger than two) variability.

The sand wave dynamics are evaluated using a 2D cross-correlation. This technique computes local morphodynamics using the horizontal translations of bedforms from bathymetrical surveys. An example of this technique is presented in Figure 3.1, for an area around an analysis location. The size of the considered area is chosen such that at least a significant part of one sand wave is covered in both surveys that are compared, considering the possible migration distance of the sand wave in the period between the surveys. It is also essential to limit the area to assess sand wave dynamics locally. The red arrow in the figure indicates the shift between the two surveys (black and blue polygons).

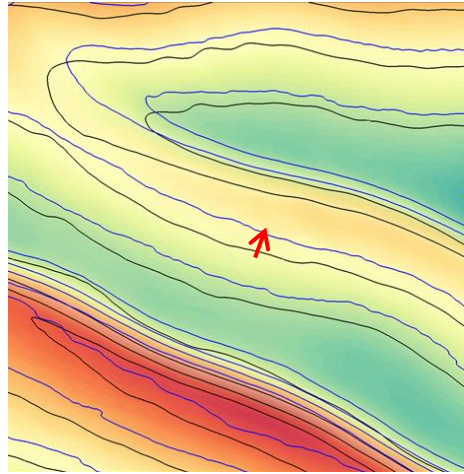


Figure 3.1: Example of the 2D cross-correlation. The red arrow indicates the highest correlation found between an older (black contour lines) and a more recent bathymetry (blue contour lines and presented as the figure background), with the arrow length indicating migration distance and the orientation migration direction.

Based on the above-presented method, the sand wave migration direction (direction of the arrow) and the sand wave migration distance (length of the arrow) are estimated for each 5x5 km block. The sand wave migration rate is derived by dividing the migrated distance, as derived from the 2D cross-correlation of two bathymetries, by the difference in measurement dates of the two bathymetries.

This analysis yielded a single value for the direction and rate per analysis location per survey comparison. It is noted that no values are found if one or both surveys contain insufficient data for the specific analysis location. Local variations over time are quantified to capture the full temporal variability of sand wave dynamics at each analysis location. More specifically, the distribution of the migration directions and rates of all the neighbouring locations within a selected distance is calculated for each analysis location. The distance is selected such that the enclosed area is small enough to maintain local information, but at the same time, it should contain enough points to determine probability distributions.

Sand waves migrating in opposite directions might cause issues when determining the spread in migration directions. Averaging the migration direction for sand waves migrating in opposite directions can result in a resulting average migration direction perpendicular to the actual direction of migration. This error can occur when determining the spread in migration directions for a single analysis location or when interpolation is performed between several analysis locations.

To cover for sand waves migrating in opposing directions, first a prevailing direction of migration is determined. Since sand waves are characterised by a steeper lee slope oriented in the direction of migration, the most prevailing direction of this lee slope is considered. All sand waves migrating roughly opposite of this direction are considered to have a negative migration rate, hence the direction of migration is flipped 180 degrees. For example, when the prevailing direction of migration is 20°N, and a sand wave is migrating with two metres per year towards 190°N this, the sand wave is considered to migrate with a rate of minus two metres per year towards 10°N.

During the processing of the entire DCS at several locations unrealistic high migration rates were calculated. These were the result of either detected sand wave crests that are no sand waves (e.g. due to dredging activities) or at the edges of surveys. To remove these outliers two

steps were taken for each 5x5 km block including the buffer zone: 1) migration rates above 100 metres per year were discarded and 2) migration rates above the 99.5% percentile were discarded.

After removal of the outliers minimum and maximum rates were determined by taking the average rates plus or minus two times the standard deviation. Including the minimum and maximum rates reduces uncertainties in the temporal evolution of sand wave migration directions and rates. The resulting bandwidths including mean values, are used to assess future and historical seabed levels.

3.4 Extrapolation of historic trends

The assessment of future and historical seabed levels consists of extrapolating historical seabed level trends. The extrapolation is performed only on the separated sand wave morphology. For the extrapolation the horizontal migration of the bedforms is used.

Bedforms for which no historical trends can be determined, such as the highly transient megaripples, are considered uncertain. It is noted that when considering historical bed levels the observed historical trends are reversed in the extrapolation. The approach is discussed in more detail below.

3.4.1 Extrapolation of horizontal migration of bedforms

The future and historical bathymetries, as well as the corresponding bed level changes for the areas with dynamic bedforms are estimated by artificially shifting the bedforms. The starting point is a bathymetry covering the full 5x5 km block including the 2.5 km buffer zone.

In most cases, the available bathymetries are measured in different years and they do not always cover the entire block with buffer. To cater for this, a composite bathymetry is created. In case multiple bathymetries are available, the most recent bathymetry is used. It is noted that the 2021 and 2022 surveys by Royal Netherlands Navy - Hydrographic Office (1980-2021) are not considered in the extrapolation (see Section 2.1).

To cover for the temporal differences in the composite bathymetry, each component is shifted using the mean migration direction and rate over this period. For example, when a composite bathymetry is created from bathymetrical surveys measured in 1992, 2005 and 2020, the bathymetries measured in 1992 and 2005 are shifted over 28 and 15 years, respectively. This results in estimated seabed levels (based on shifting the 1992 and 2005 bathymetries) and a measured most recent bathymetry (2020). By combining these with the measured most recent bathymetry prevailing, a composite bathymetry is created and used as starting point in the data extrapolation. This is further explained in the extrapolation of an individual scenario below.

The artificial shifting of the sand waves is done with the aid of the determined migration directions and rates bandwidths. These bandwidths contain a lower bound, mean and upper bound for both the migration direction and rate, resulting in 9 possible displacements fully describing the temporal variation in bedform dynamics.

First, the bandwidths for all analysis locations are interpolated, resulting in bandwidths for the entire grid used in the analysis. When combining the migration directions and rates, for every grid point, the yearly shift in x and y directions is calculated. Based on the bandwidths, two approaches for the extrapolation were used, which are further explained below.

3.4.2 Extrapolation of an individual scenario

The first approach for extrapolation consists of shifting the composite bathymetry using a single scenario, i.e. a combination of a single migration direction and rate per grid point. This approach is used to determine best-estimate scenarios consisting of the mean migration direction and rate extrapolation. An example of this approach is shown in Figure 3.2, where per grid point (black dots) the displacement in x and y directions is visualised using an arrow. This arrow indicates the direction of migration and the total displacement in that direction. This total displacement is calculated by multiplying the shift in x and y directions and the number of years for which the extrapolation is performed.

This approach results in grid points containing a certain bed level being extrapolated to a new x and y location. The resulting x, y and z locations are ultimately interpolated back to the original x and y locations resulting in an estimated seabed level for a specific scenario for a specific year which can be both below and above the most recent measured seabed level.

Using this method, an estimation of the bathymetry is made for 2022 and from 2025 till 2050 every five years.

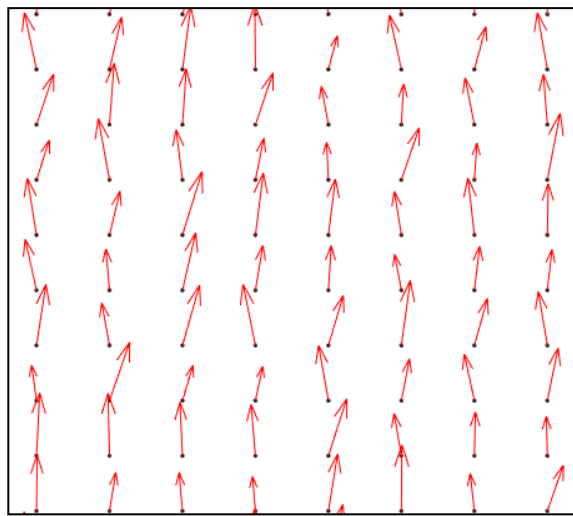


Figure 3.2: Example of shifts per grid point (black dots) with the direction of the arrow indicating migration direction and the length of the arrow the migration rate (for illustration purposes, the grid resolution and scale of migration rates is not indicated in the figure).

3.4.3 Extrapolation of cumulative seabed levels

The second approach for extrapolation consists of extrapolating the full bandwidth of shifts in x and y directions over a specific period. This is relevant when the lower and upper envelope of predicted seabed levels is requested over a specific period. Examples are for the full lifetime of the wind farm relevant for determining initial cable burial depths.

In this approach, a polygon is drawn for each grid point which covers the area between the lower and upper bound migration direction and between the analysed grid point and maximum displacement away from the grid point (calculated by multiplying the upper bound migration rate and the number of years for which the extrapolation is performed). This polygon describes the area where the analysed grid point (thick black dot) can be in the period assessed (e.g. 1945 till 2022).

An example of this approach is shown in the left panel of Figure 3.3, with the drawn polygon shaded blue and the analysed grid point as a thick black dot. This approach is performed for all grid points resulting in the lower or upper envelope of the predicted seabed levels for the period assessed.

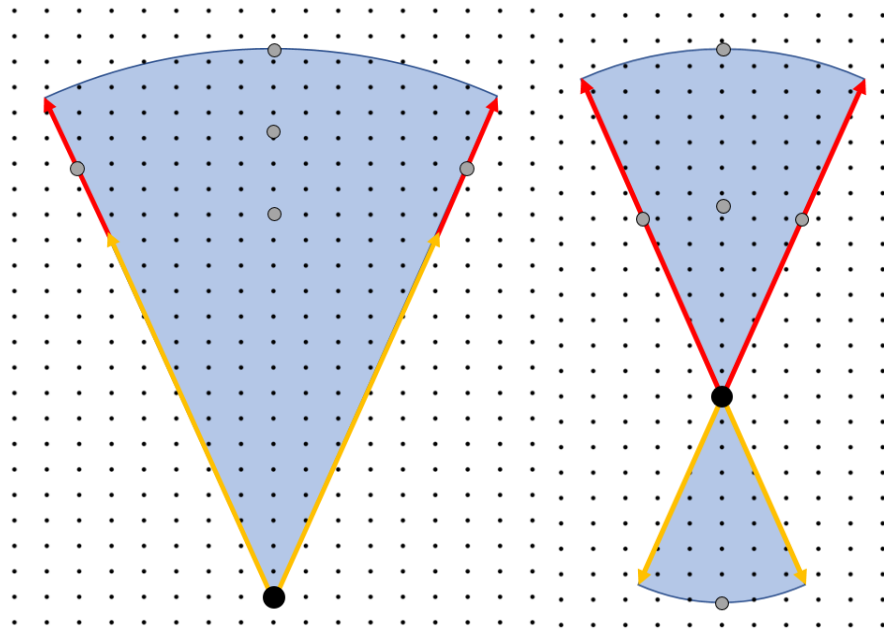


Figure 3.3: Example of the extrapolation approach indicating a case where both the lower and upper bound migration direction have the same sign (left panel) and the case where opposite migration is observed (right panel). The analysed grid point is indicated by a thick black dot whereas the other grid points are indicated by smaller black dots. The lower bound displacement for the lower and upper bound migration directions are indicated with orange arrows, the upper bound displacements by the red arrows. The area between the analysed grid point and the upper bound displacement (left panel) and between the lower and upper bound displacements (right panel) between the lower and upper bound migration directions describe where the analysed grid point can be assessed over the considered period (e.g. over the lifetime of the wind farm). All other scenarios, e.g. mean direction and mean displacement, are indicated by grey dots and always fall inside the blue polygon.

3.5 Sources of uncertainties

In assessing the future and historical seabed levels, various sources of uncertainty have to be taken into account, both upward and downward, i.e. upward uncertainties are applied to the upper envelopes of extrapolated seabed levels, whereas the downward uncertainties are applied to the lower envelopes of extrapolated seabed levels. For the purpose of identifying depth of burial of (CE) objects only the downward uncertainty is relevant. The primary sources of uncertainty in a data-driven morphological analysis based on measured bathymetrical data are due to:

- i) Data collection and differences in the collection of data;
- ii) The finite and limited grid resolution;
- iii) Sand wave reshaping; and
- iv) Megaripple dimensions and large-scale seabed variations.

3.5.1 i) Uncertainty due to data collection and differences in the collection of data

The bathymetries used are collected using Multi Beam Echo Sounding (MBES) or Single Beam Echo Sounding (SBES) (most often surveys performed before 2000). The method used introduces uncertainties, larger for the SBES systems but still relevant for the MBES systems.

For the more recent data, the vertical uncertainty is 0.20 m, which is used in this study. Data provided by Rijkswaterstaat, TenneT and RVO will all meet this uncertainty, for the NLHO surveys this might be higher, especially for the older surveys. However, since only the most recent surveys are used for the extrapolation, the uncertainty of 0.20 m shall be a reasonable value for major part of the DCS studied.

3.5.2 ii) Uncertainty due to the finite and limited grid resolution

Furthermore, the bathymetrical data is gridded (either by Deltares or the provider of the data) to a raster of data points with a fixed resolution in x- and y-direction to be used in further analysis. Although a larger grid cell size can still be used to capture constant slopes accurately, it is expected to introduce errors in the bathymetry schematisation at the areas with changes in the slopes and larger slopes such as the troughs and peaks of the sand waves and megaripples.

Since many different data sources were used, all with different grid resolution and processing, for this study one fixed value of 0.20 m is used for this uncertainty. This is based on expert experience within other morphodynamic studies.

3.5.3 iii) Uncertainty due to sand wave reshaping

To account for changes in the shape of the sand waves, an uncertainty of 0.25 m at the crest and 0.15 m at the rest of the sand wave is applied. This value is based on expert judgement and results from the assessments for IJmuiden Ver and Hollandse Kust (west) (Deltares, 2020b, 2023).

3.5.4 vi) Uncertainty in megaripple dimensions

For smaller-scale bedforms for which no historical trends could be determined, the inclusion in the extrapolation of seabed levels is treated in three parts. This is when bedforms are highly transient or subject to changes under extreme events.

First, the spatial distribution of megaripples is included in predicting future bathymetries as the unfiltered mobile bathymetry is used in the extrapolation. Second, the uncertainty induced from using a coarser grid in the analysis compared to the resolution of the measured data is quantified and included as described under ii).

Lastly, a significant limitation is related to the temporal variation of the megaripples. The bathymetrical surveys are a snapshot recording at a single point, while the occurrence and dimensions of the smaller-scale bedforms such as megaripples may vary in time. Therefore, a fixed value is taken up in the uncertainty band to cover this temporal variation. For the determination of the lowest seabed levels an uncertainty of 0.10 m is used.

3.5.5 Summary of uncertainties

The total uncertainty band that is applied on the extrapolated future and historical seabed levels is based on the above denoted sources has similar components for the downward and upward direction and contains the following:

- i) Survey inaccuracy: 0.20 m;
- ii) Uncertainty due to grids: 0.20 m;
- iii) Sand wave shape: 0.25 m at the crest and 0.15 m at the rest of the sand wave; and
- iv) Megaripple uncertainties: 0.10 m.

3.6 Future and historic seabed level assessment

The historical trends, measured bathymetry, extrapolation approach and uncertainties are combined for future and historical seabed levels. The method consists of the following components, in which the component for smaller-scale bedforms such as megaripples is taken up in the sand waves and uncertainties:

- Large-scale seabed;
- Sand waves; and
- Uncertainties.

An example of such an extrapolation is shown in Figure 3.4, indicating bed level predictions including bandwidths of different contributors. The figure presents sand wave variability concerning the underlying large-scale seabed. The upward and downward variability due to uncertainties is included concerning the sand wave variability. The influence of future infrastructure, such as foundations and scour protections, is not considered in the presented bandwidths because of uncertainties. This influence can, however, be significant on local seabed level variations.

The evolution over time of both the lowest and highest seabed levels is not a linear process. A given location can expect both seabed lowering and seabed rise over a considered period in case a sand wave is migrating over. In this case, for the first years, the highest seabed level may rise, as the sand wave crest migrates towards a certain location and remains constant for the remainder of the period. i.e. no new highest seabed level is observed as the sand wave crest is migrating away from the given location. In case a higher sand wave migrates past this specific location a new highest seabed level is recorded.

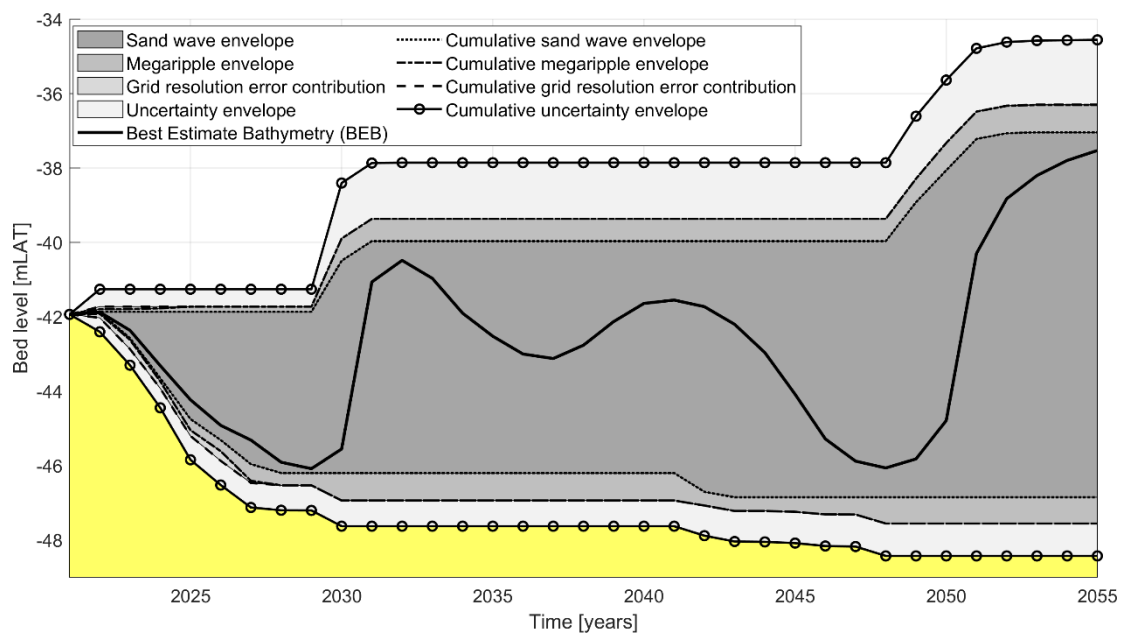


Figure 3.4: Example of future bed level predictions (i.e., bed level as function of the time) for a fixed location in the DCS including bandwidths of different contributors. Note that the predictions are valid for a single exact location and do not represent an average over a defined area. To assess the burial depth of (CE) objects only the lower boundaries are relevant. NB: the lower end of the envelope represents the LSBL, not the BSBL or HSBL (see Section 3.6.1).

3.6.1 Future seabed levels

Future seabed levels are determined to have an indication of (CE) objects burial depth in the future, every 5 years between 2025 and 2050. Each of the future seabed levels has several

components, including seabed trends and uncertainties. The lowest and highest seabed levels are expressed as cumulative, being the upper and lower envelope of seabed predictions over a given period.

The following future seabed levels for various years and periods are considered:

- Best-Estimate Bathymetry (BEB): an estimate of future seabed levels using the mean migration rate and direction;
- Low-Estimate of the lowest SeaBed Level (LSBL): the predicted lowest minimum seabed level for a specific period incorporating uncertainties (indicated by the lower end of the envelope presented in Figure 3.4);
- Best-Estimate of the lowest SeaBed Level (BSBL): the predicted minimum seabed level for a specific period incorporating uncertainties, based on mean migration rates;
- High-Estimate of the lowest SeaBed Level (HSBL): the predicted highest minimum seabed level for a specific period.

The BEB only provides a very rough indication of the possible seabed development during the lifetime of the wind farm and should not be treated as a firm design parameter. For design, the LSBL and HSBL provide better information (maximum expected potential seabed level variations at each grid point). However, the BEB does provide a valuable estimate of the seabed to compute the most probable O&M costs (e.g. related to expected cable re-burial length).

The seabed close to possible scour protections and cable crossings will most likely lower more than the LSBL levels because of local scour and edge scour. Buried electricity cables that cause no flow disturbance will only incur this scour effect if the cables are exposed on the seabed.

3.6.2 Historical seabed levels

Historical seabed levels are presented to indicate the lowest seabed levels over the period since World War II until the most recent measurement of each 5x5 km block. It is assumed that relatively small objects, such as CE, cause no/negligible flow disturbance themselves and, therefore, will only cause local scour that can result in partial settlement of the object. These objects will, however, not affect the processes responsible for sand wave dynamics. Therefore, they will experience coverage if a sand wave crest passes over.

After the initial impact, a CE in a sandy environment might partially self-bury due to scouring (Menzel et al., 2022). It is also expected to experience burial and/or exposure due to bedform migration or large-scale seabed level changes. Under the effect of these morphodynamic processes the CE is assumed to stay at the lowest seabed level it has experienced between 1945 and now. Quantification of initial penetration of CE into the seabed is not part of the scope of this study. If significant penetration occurred during impact, then at locations that mainly experienced seabed rising the actual vertical level of the CE may be overestimated. Additionally, potential movement of an exposed CE under hydrodynamic or wave forcing is not assessed.

The following historical seabed levels are determined. It must be noted that historical trends are reversed when determining the historical seabed levels. Furthermore, the historical seabed levels are the minimum value of all measured high-resolution bathymetries between 1945 and now and the hindcasts:

- Lowest Object Level (LOL): the predicted minimum seabed level since World War II;
- Best-Estimate Object Level (BEOL): the predicted best-estimate lowest seabed level since World War II;
- Highest Object Level (HOL): the predicted estimation of the maximum lowest seabed level since World War II.

When comparing the levels to Figure 3.3, the polygon covers the area between the lower and upper bound migration direction and between the analysed grid point and displacement away from the grid point (calculated by multiplying the negative (because of a hindcast) lower bound (HOL), best-estimate (BEOL) and upper bound (LOL) migration rate and the number of years for which the extrapolation is performed).

The estimated CE burial depth is obtained by taking the minimum of the historical and future results and subtracting this from the estimated bathymetries (BEB's).

3.7 Expected coverage of objects

The historic and future seabed levels are used to compute the thickness above an expected CE. The minimum, mean and maximum thickness for a specific year (Y) are computed according to the following equations:

- Minimum thickness above an expected CE for year Y = $BEB_Y - \min(HOL_{1945}, HSBL_Y)$
- Mean thickness above an expected CE for year Y = $BEB_Y - \min(BOEL_{1945}, BSBL_Y)$
- Maximum thickness above an expected CE for year Y = $BEB_Y - \min(LOL_{1945}, LSBL_Y)$

3.8 Quality checks

During the different steps of the analysis several checks were performed. The most important parameters to have a good end result are the correct separation of the large-scale bathymetry from the sand waves and the correct migration rates.

Due to the highly varying morphology of the DCS an automated method to do a quality check of the results was not feasible. Instead, output for all analysis steps was analysed and checked for outliers. For this purpose visual checks on specific areas of interest (e.g. with high mobility or limited number of surveys) were carried out. Outliers were corrected for by implementing exceptions to the methodology. Very specific outliers, for example a single outlier for migration rates in an area with very limited mobility, were corrected individually.

Also, the large variety and number of datasets used in this study resulted in many unexpected artefacts which were best discoverable by visual inspection, because they resulted in abrupt changes between different areas in the DCS. In this section, the most important checks on the separation of the bathymetries and the migration rates are shown.

3.8.1 Splitting bathymetries

The settings for the automated 2D filtering used in this study were based on the characteristic dimensions of sand waves. To check the result, 2D Fourier plots (power plots) of the separated large-scale bathymetry, sand wave field and megaripple field were made (Figure 3.5). These plots show how much energy is present for different wave lengths. In the large-scale bathymetry little energy should be present in the sand wave range, while in the sand wave bathymetry little energy should be present in the larger scale range (boundary is roughly indicated by the dashed lines). As can be seen, the *presence* of the sand waves is also correlated to the large-scale morphology, with large heights on the tops of the sand banks and smaller heights or absent in the trough or the westside slope. This causes the larger wave lengths on the y-axis in the power plot. For the megaripples a similar effect can be seen.

Also, visual checks of areas with different morphology are made (Figure 3.6). In the middle figure there are still rhythmic patterns visible in the large-scale bathymetry. This is caused by groups of sand waves with a length of about 2 km. These rhythmic patterns are thus much longer than sand waves, so are correctly part of the large-scale bathymetry. Also, in the other areas the separation shows good results. Some effects can be seen from the survey boundaries, which are clearer because the survey date might differ.

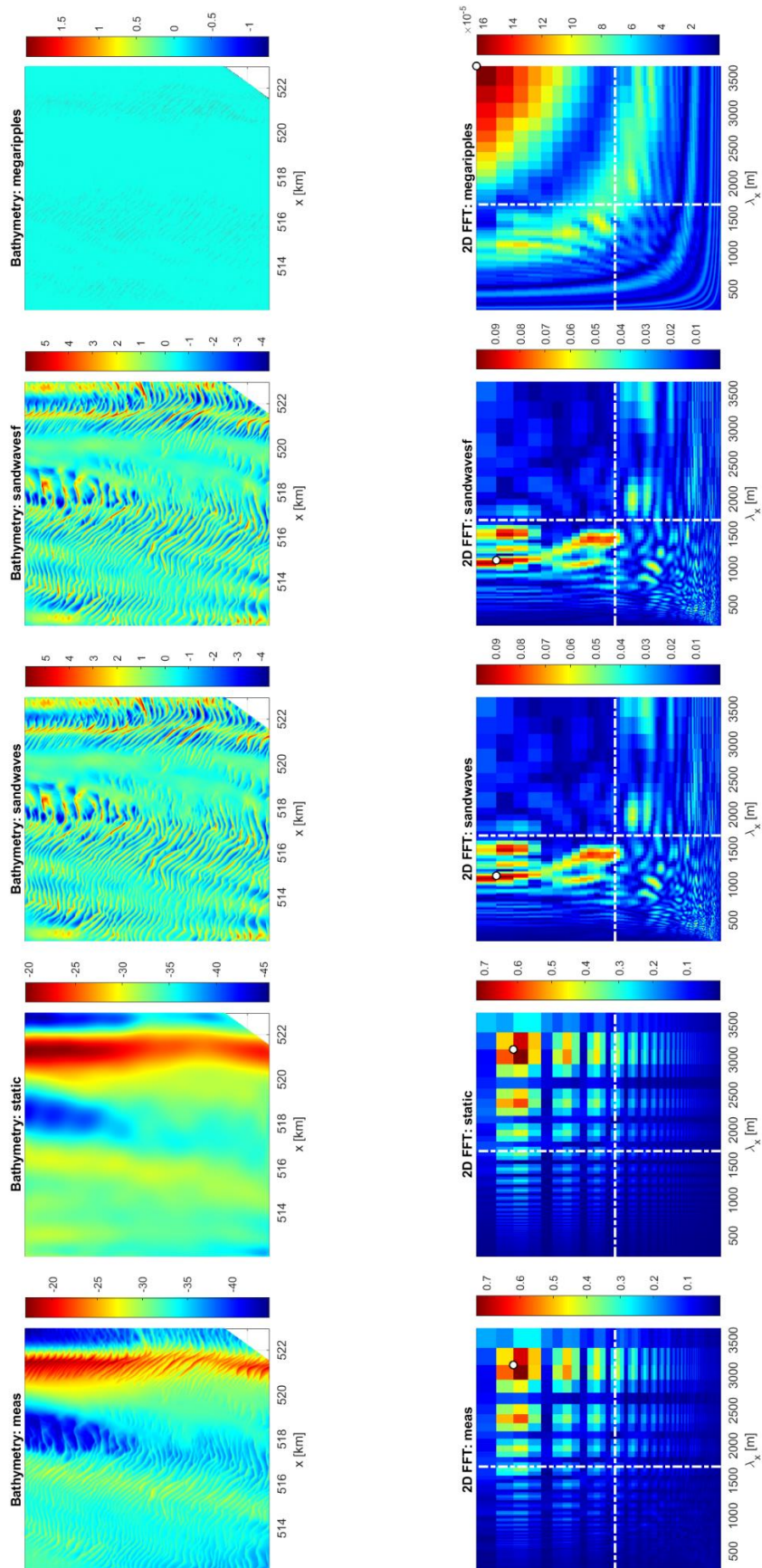


Figure 3.5: Separated bathymetries and corresponding 2D Fourier plots. White dashed lines indicate the boundary of sand waves and larger scale forms, both in x and y direction

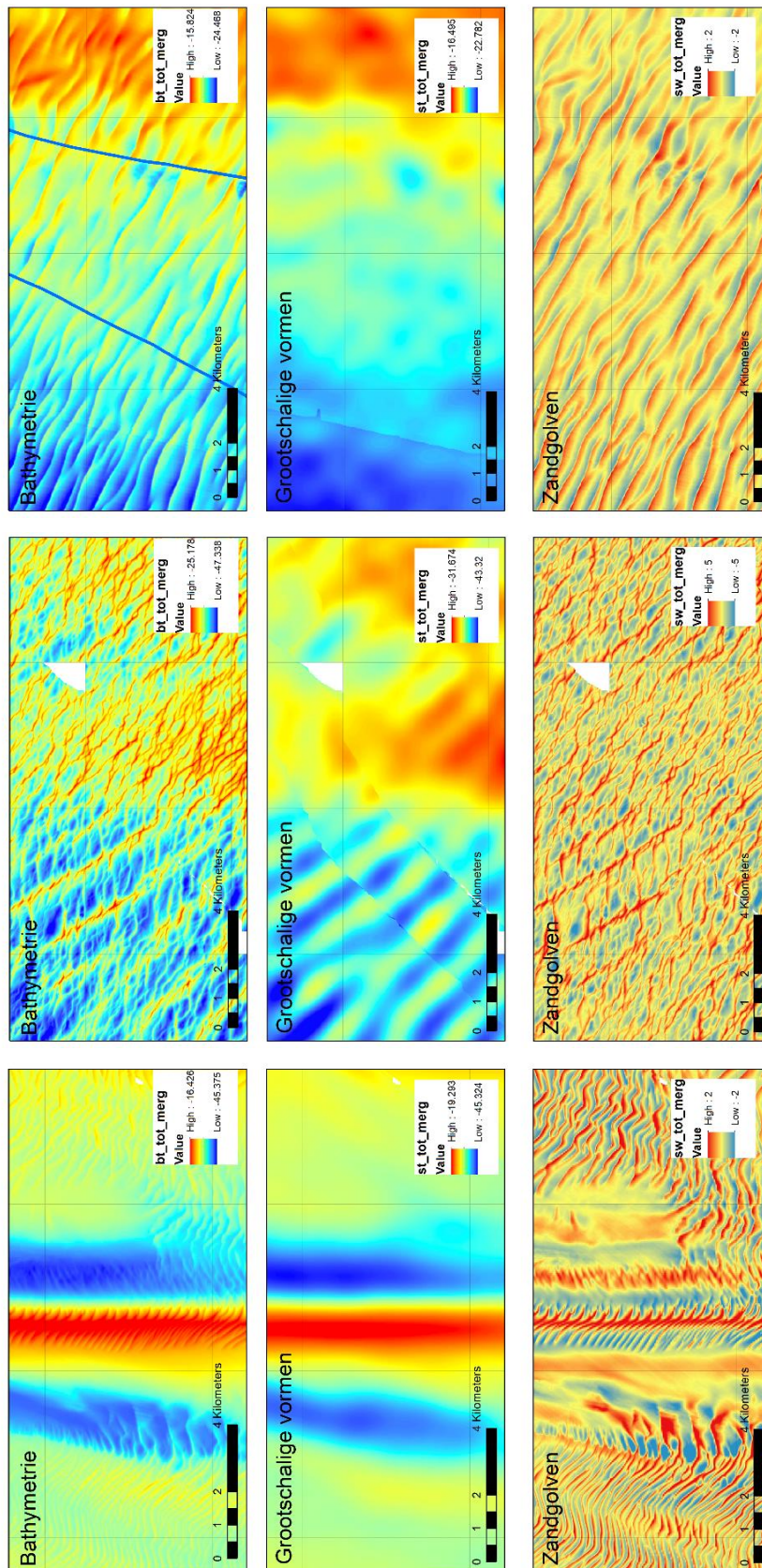


Figure 3.6: Examples of splitting of bedforms in three areas with very different morphology: 2D sand waves with long, straight crests (upper), 3D sand waves with many bifurcations of the crests (middle) and sand waves on large sand banks (lower). Blue lines in upper left panel are the 12 nautical mile and NAP -20 m line.

3.8.2 Migration rates

To obtain correct migration rates the removal of outliers is important. Plots with different cut-off settings were made for a number of 5x5 km blocks where high rates were found in the results (Figure 3.7). For example, outliers up to 40 m/year were found whilst the surrounding area indicated values up to about 2 m/year with a more realistic spatial distribution. Introducing cut-off settings, with in particular values in an area that exceed the magnitude 99.5% of all migration rates found, reduced the number of outliers significantly.

Outliers also occurred due to erroneous surveys. Inspections of the minimum, mean and maximum migration rates yielded several areas for which migration rates were found to deviate from surrounding areas. Especially in case abrupt changes were found, e.g. along lines delineating extents of surveys, these areas were highlighted as erroneous.

Wrong migration rates also occurred due to erroneous surveys. These were identified by manually inspecting the minimum, mean and maximum migration rates in ArcGIS. An abrupt change in migration rates along a straight line was for many locations the indication for a bad survey (Figure 3.8). To prevent unnecessary discarding of valuable survey data – the number of surveys in time is limited for many locations on the DCS – histograms of the migration rates per survey-combination were made (Figure 3.9). In the case of a ‘bad’ survey, all combinations with that specific survey showed deviating rates compared to all the other combinations. At some locations only some combinations showed deviating rates in one block, while in the neighbouring blocks this was not the case. In such cases the survey was not taken out of the set, but the area was marked as unreliable.

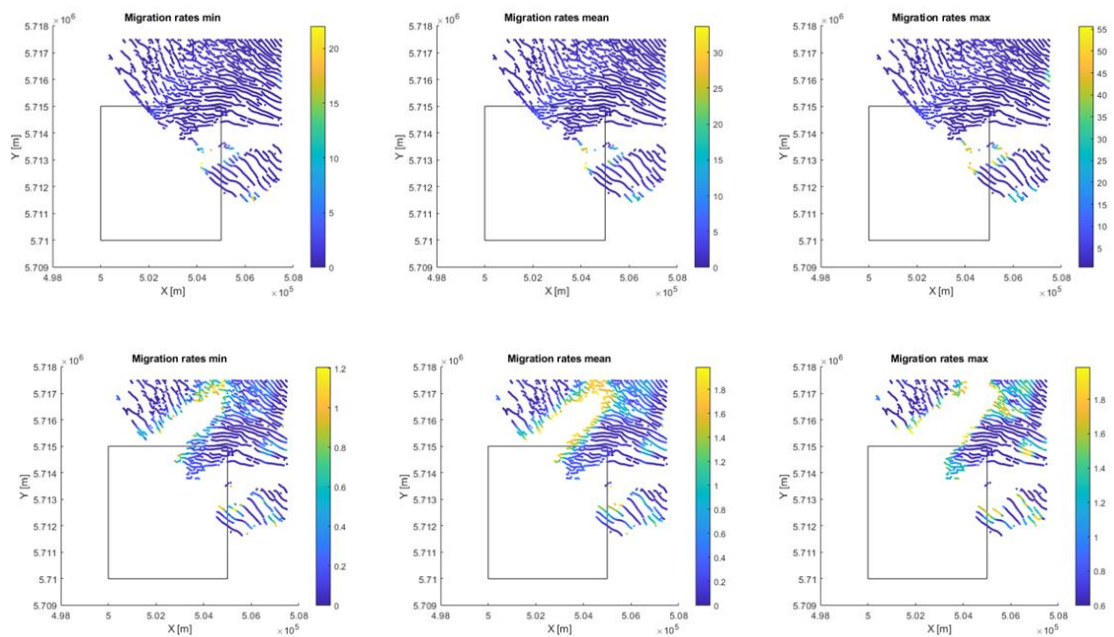


Figure 3.7: Original migration rates (upper) and migration rates after removal of rates above the 99.5% percentile (lower). NB: colour scale differs per subplot.

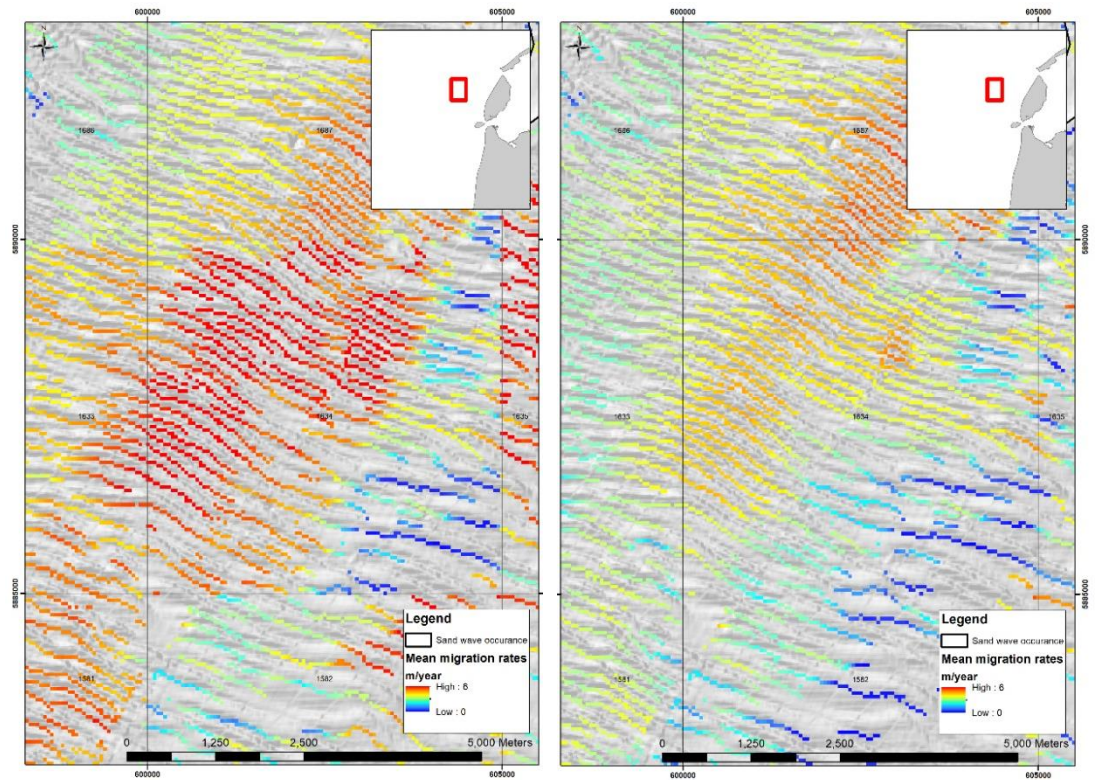


Figure 3.8: Mean migration rates in sand wave field offshore Texel including surveys sid386 and sid1093 (left) and after removal of those surveys (right).

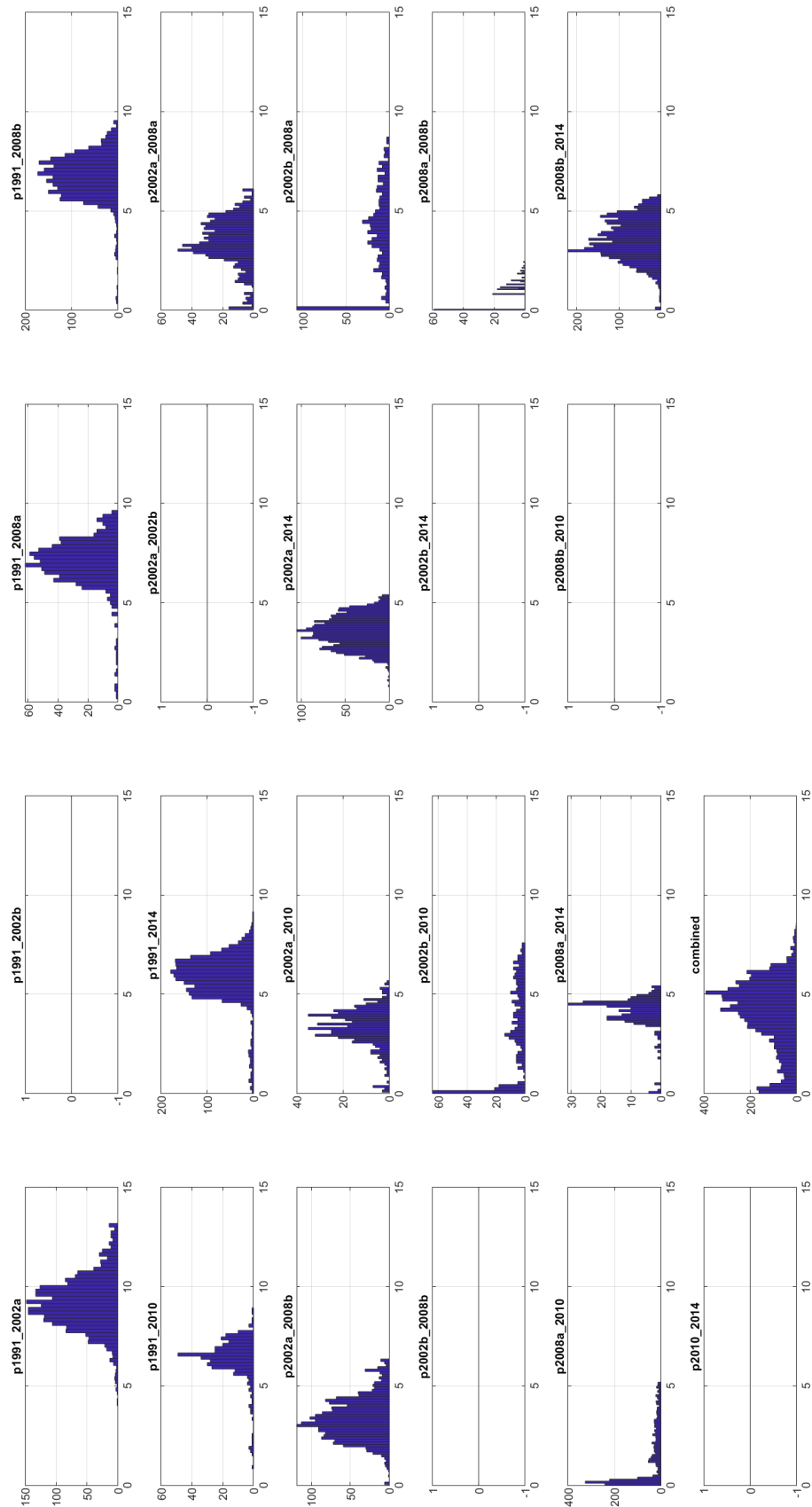


Figure 3.9: Histograms of migration rates for different survey combinations (empty plots are combinations of surveys that don't overlap in the analysed area).

4 Results

In this chapter, maps are presented with the main results from the separation of bedforms (Section 4.1), the determined migration rates (Section 4.2), the extrapolated future and historic seabed levels (Section 4.3), and the expected coverage of an expected CE (Section 4.4) and the Best-estimate of the seabed lowering towards 2050. Finally, based on the results a map has been made to indicate the quality of the results (Section 4.5).

4.1 Separated bathymetry

Figure 4.1 gives an overview of the most recent bathymetry used in this analysis for the sand wave area of the DCS. Figure 4.2 and Figure 4.3 show the large-scale seabed and sand wave field for the same area.

Figure 4.1 depicts the bathymetry, which shows that the deepest water is found along the southwestern edge of the sand wave region. As expected, the water depth gradually decreases towards the coast. Figure 4.2 clearly displays the north and south tidal sand banks, with the northern banks running from north to south and the southern banks facing the coast of Zeeland. In the southwestern part of the sand wave area, long bed waves with wavelengths of around 1.5 km can be seen. These rhythmic features are larger than sand waves, which typically have lengths ranging from 100 to 1000 m.

Figure 4.3 displays the varying patterns of sand wave characteristics, such as length, height, orientation, and shape, across the DCS. Despite these patterns, the relatively small size of the bed features makes them difficult to observe. Generally, the length of the sand waves increases in the north and their height decreases towards the east. The southwestern region is characterized by braided sand waves, while the northeast has more regular sand waves. Near the coast, the sand waves become elongated with reduced heights and increased lengths. On average, the sand waves face 30 degrees clockwise from North, though the angle may vary due to larger features. On the northern banks, the sand waves are mainly oriented north on the seaward flank and slightly to the south on the inland flank. On the southern banks, the opposite effect is observed with northward-directed sand waves on the inland side and southward-directed waves on the seaward side.

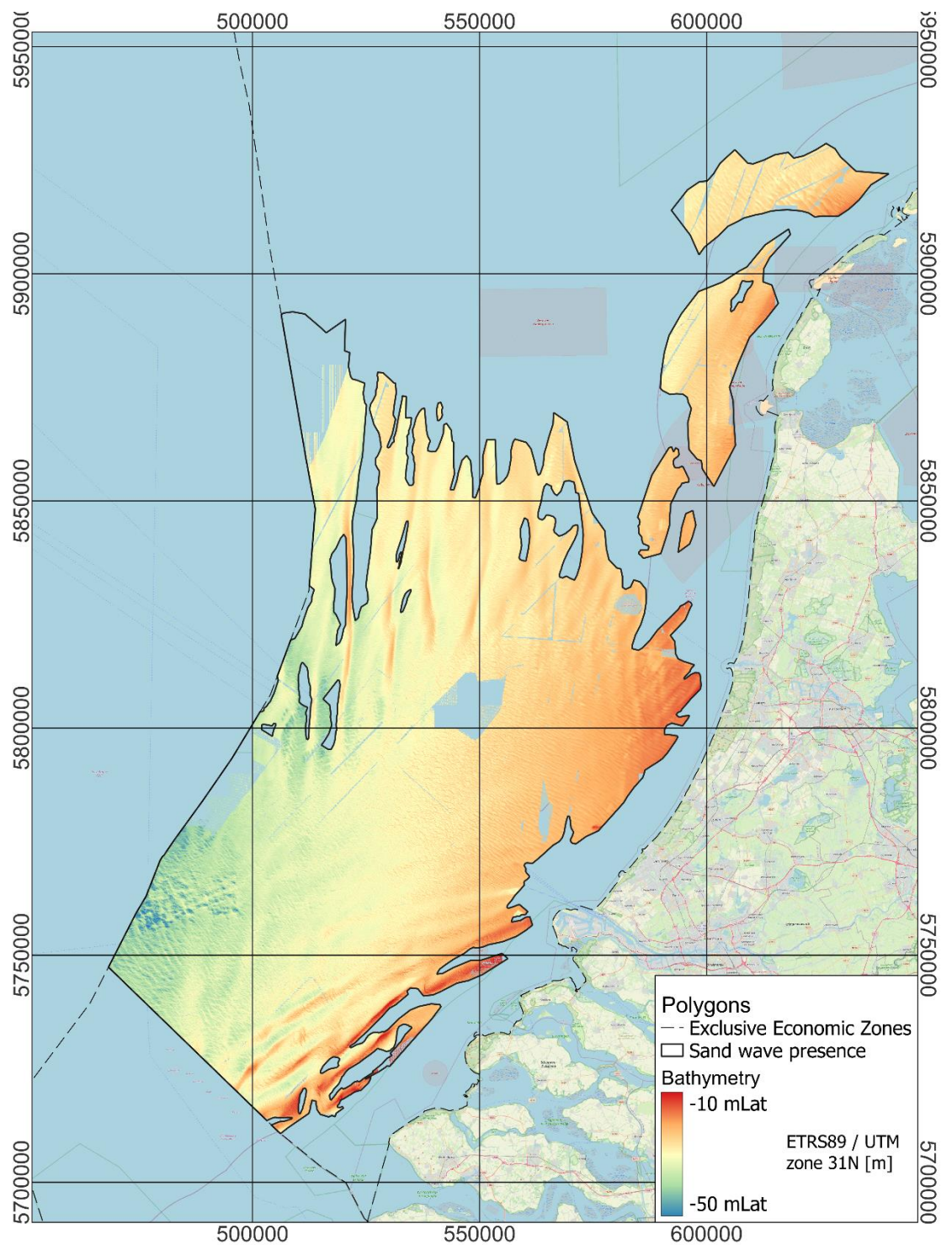


Figure 4.1: Sand wave area with the most recent bathymetry used in the analysis

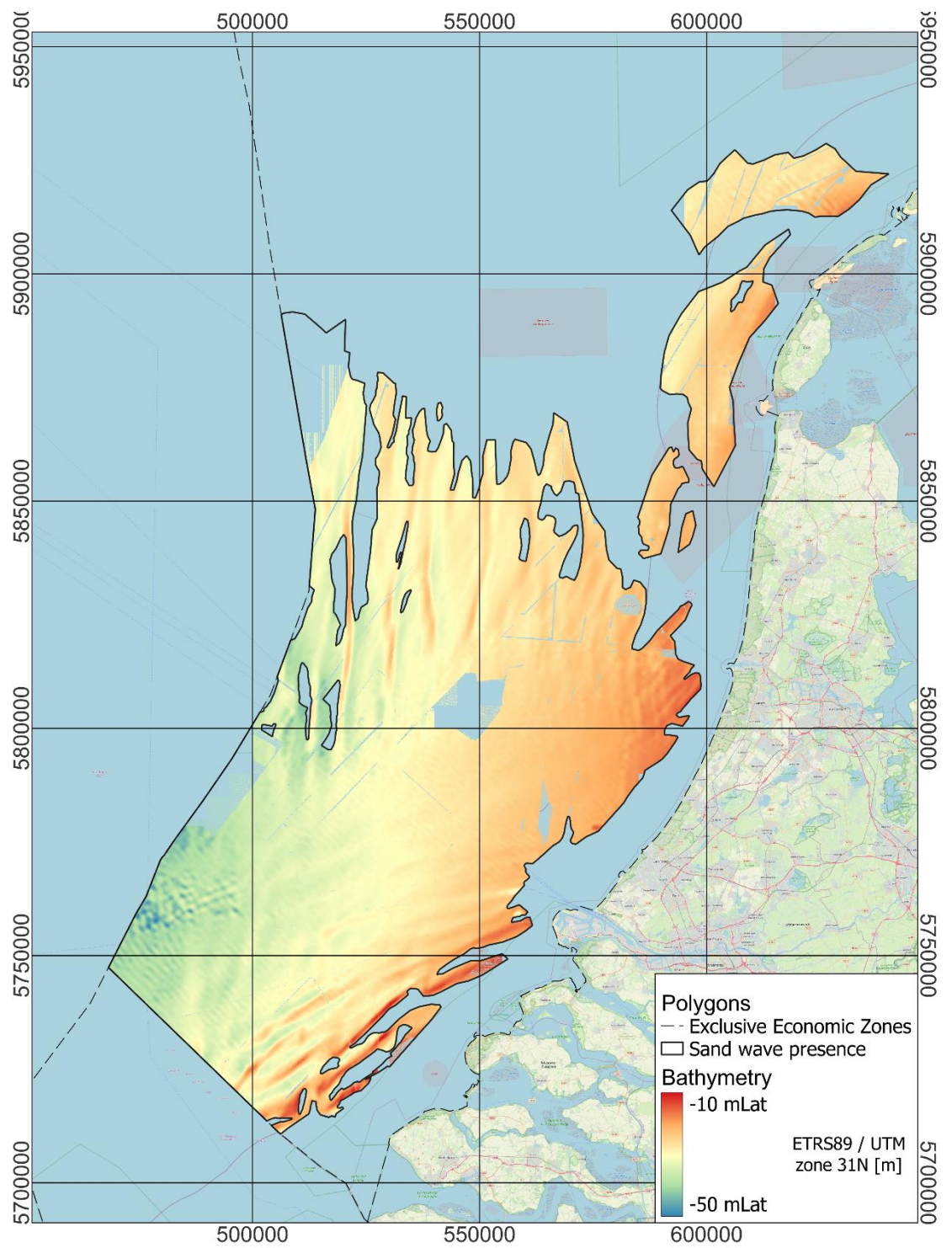


Figure 4.2: Sand wave area with the separated large-scale seabed morphology

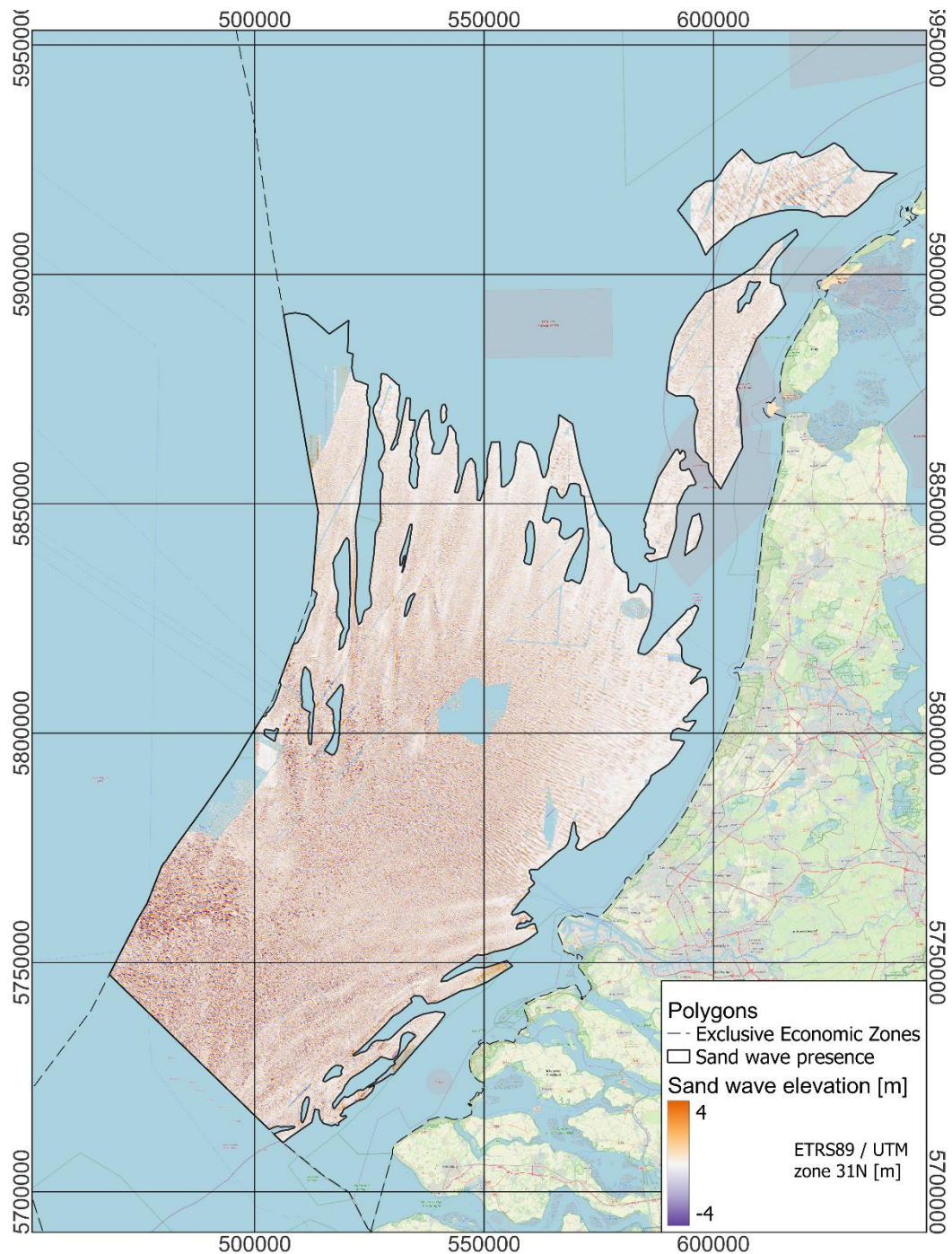


Figure 4.3: Sand wave area with the separated sand wave field.

4.2 Migration rates and directions

The mean migration rates and directions for the sand wave area of the DCS are depicted in Figure 4.4 and Figure 4.5, respectively. To illustrate the spatial variability, the migration rates and direction are interpolated to cover the sand wave area. Figure 4.4 and Figure 4.5 show that sand waves in the DCS generally demonstrates migration to the north-east, and locally to the south-west, with rates ranging from 0 to 3 m/year. In the south-western part of the sand wave area, migration rates are low. Meanwhile, the north-eastern region shows higher migration rates which gradually increase closer to the Wadden Islands. The highest migration rates are found near coastal areas and on top of sand banks, which significantly redirect the migration direction of the sand waves in several areas.

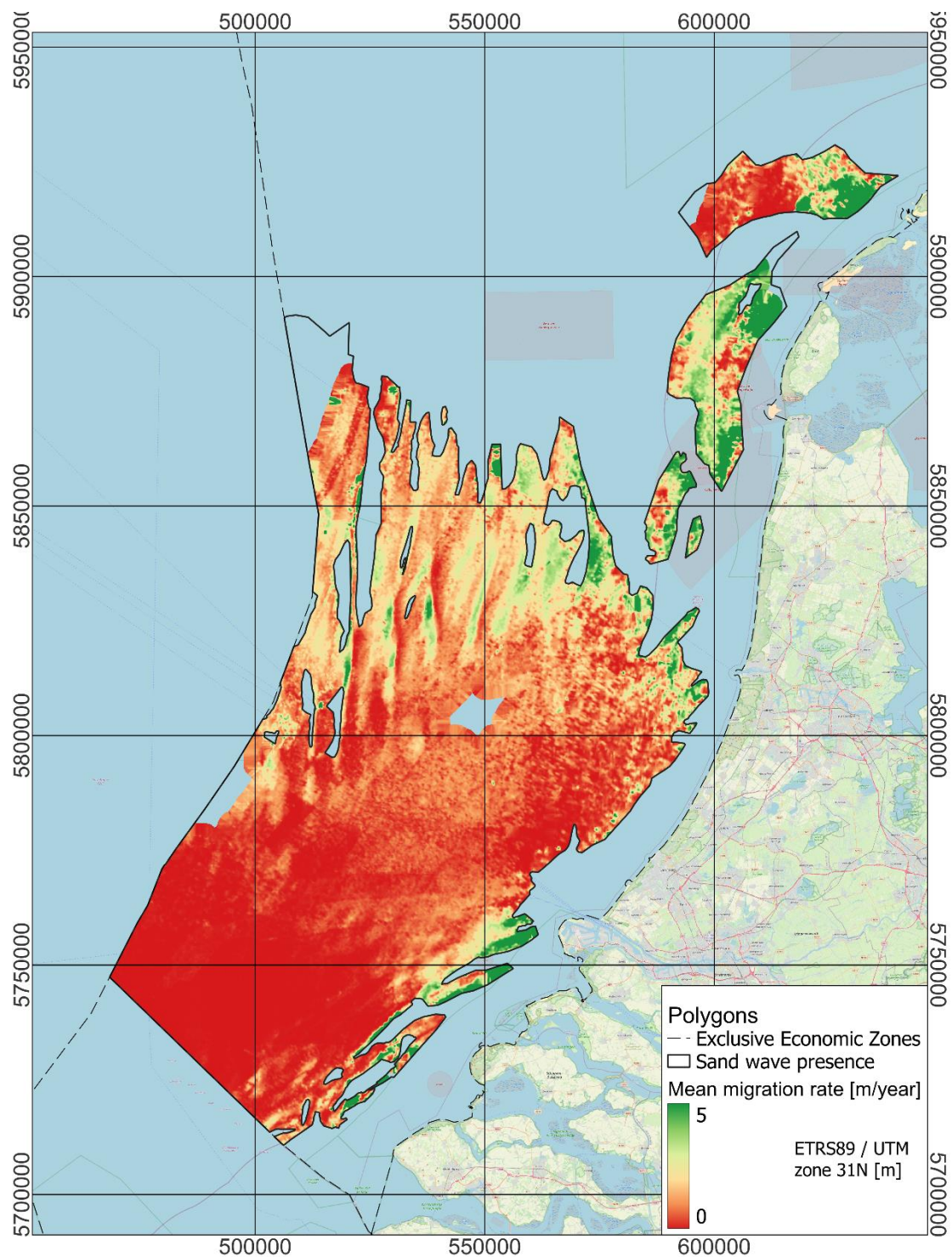


Figure 4.4: Sand wave area with the mean migration rates.

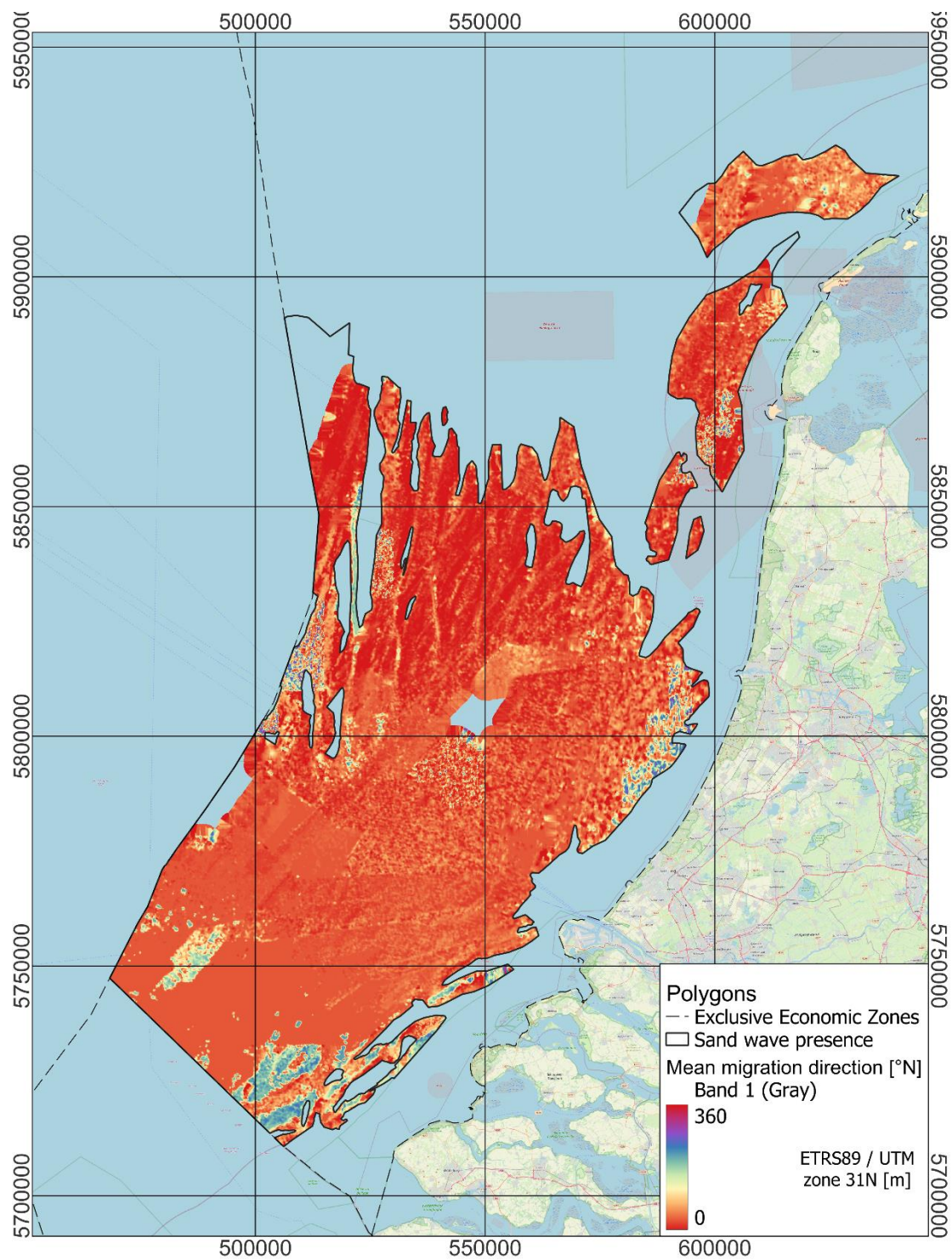


Figure 4.5: Sand wave area with the mean migration directions (in degrees with respect to North).

4.3 Historic and future seabed levels

For the sake of brevity, this report only contains figures for the BEB (Best-Estimate Bathymetry) for 2022, the BSBL (Best-Estimate of the Lowest SeaBed Level) for 2050 and BEOL (Best-Estimate Object Level) for 1945. Other levels are included in the accompanying data-package. The difference between the BEB for 2022, the BSBL for 2050 and the BEOL for 1945 with the most recent measured seabed are shown in Figure 4.6 to Figure 4.8, respectively. On this scale these bathymetries seem to be similar. However, locally differences are mainly observed around sand wave crests. These local differences are more clearly visible in the maps that show the estimated coverage of objects (see Section 4.4).

The BEB should have, on average, the smallest overall error; when compared to the actual 2022 bathymetry the BEB₂₀₂₂ is expected to have the smallest area-averaged total difference. At specific locations it may differ significantly, but the observed differences do not exceed the limits provided by the LSBL and HSBL given that the original assumptions for this analysis are satisfied.

It is assumed that relatively small objects, such as CE, cause no/negligible flow disturbance themselves and, therefore, will only cause local scour that can result in partial settlement of the object. These objects will, however, not affect the processes responsible for sand wave dynamics. Therefore, they will experience coverage if a sand wave crest passes over.

After the initial impact, a CE in a sandy environment might partially self-bury due to scouring (Menzel et al., 2022). It is also expected to experience burial and/or exposure due to bedform migration or large-scale seabed level changes. Under the effect of these morphodynamic processes the CE is assumed to stay at the lowest seabed level it has experienced between 1945 and now. Quantification of initial penetration of CE into the seabed is not part of the scope of this study. If significant penetration occurred during impact, then at locations that mainly experienced seabed rising the actual vertical level of the CE may be overestimated. Additionally, potential movement of an exposed CE under hydrodynamic or wave forcing is not assessed.

The BEB only provides a very rough indication of the possible seabed development during the lifetime of the wind farm and should not be treated as a firm design parameter. For design, the LSBL and HSBL provide better information (maximum expected potential seabed level variations at each grid point). However, the BEB does provide a valuable estimate of the seabed to compute the most probable O&M costs (e.g. related to expected cable re-burial length).

The seabed close to possible scour protections and cable crossings will most likely lower more than the LOL and LSBL levels because of local scour and edge scour. Buried electricity cables that cause no flow disturbance will only incur this scour effect if the cables are exposed on the seabed.

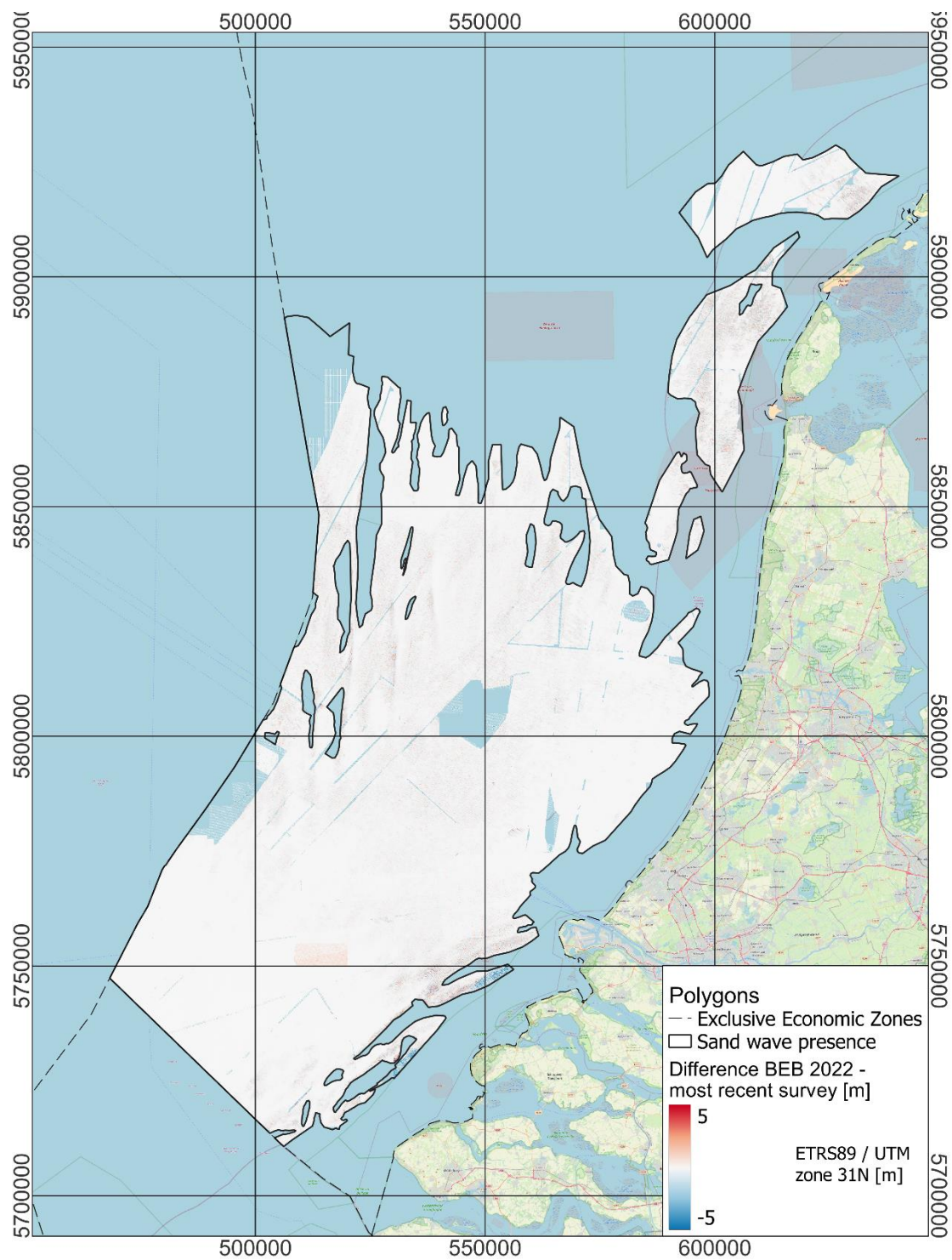


Figure 4.6: Sand wave area with the difference between the BEB for 2022 and the most recent measured seabed.

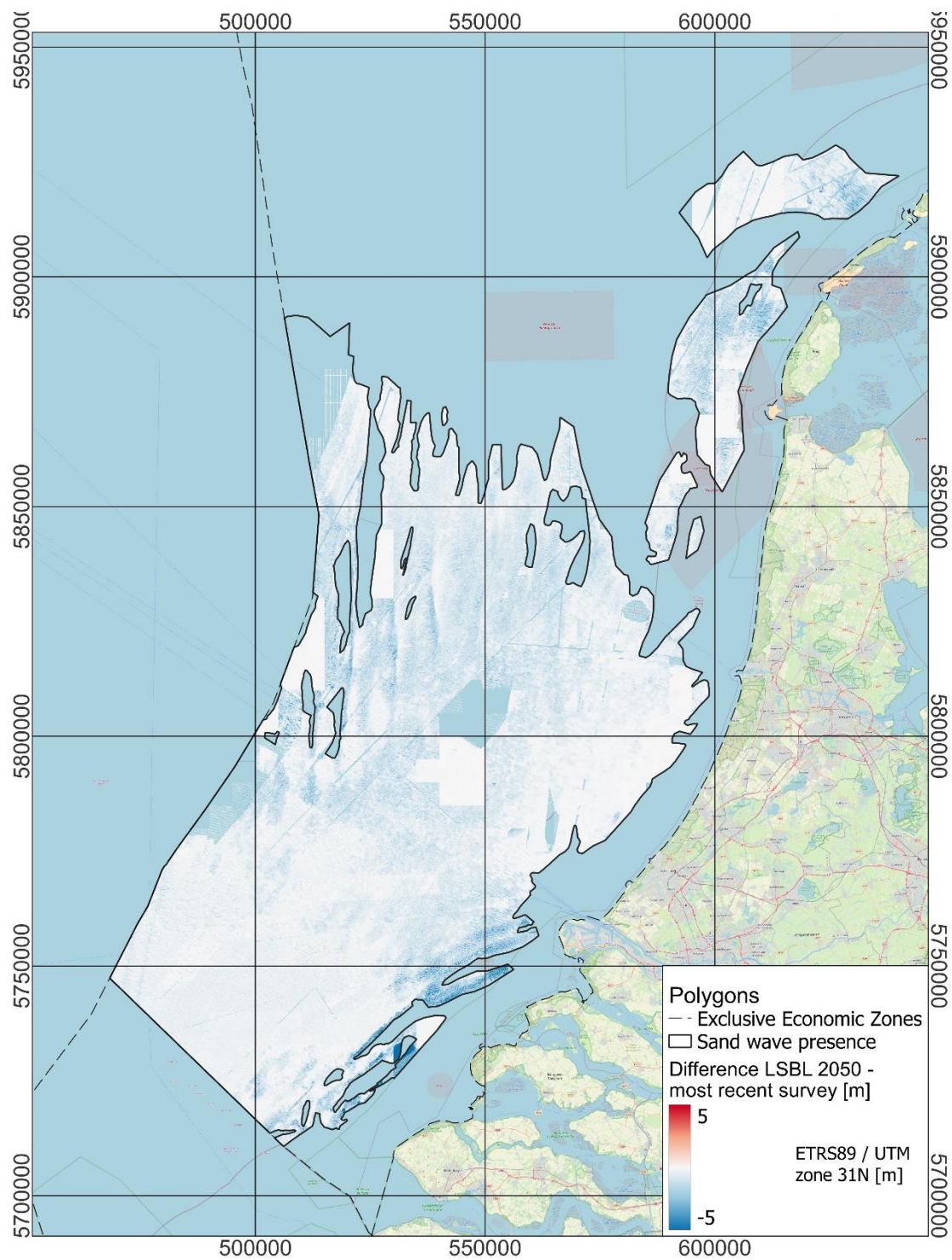


Figure 4.7: Sand wave area with the difference between the BSBL for 2050 and the most recent measured seabed.

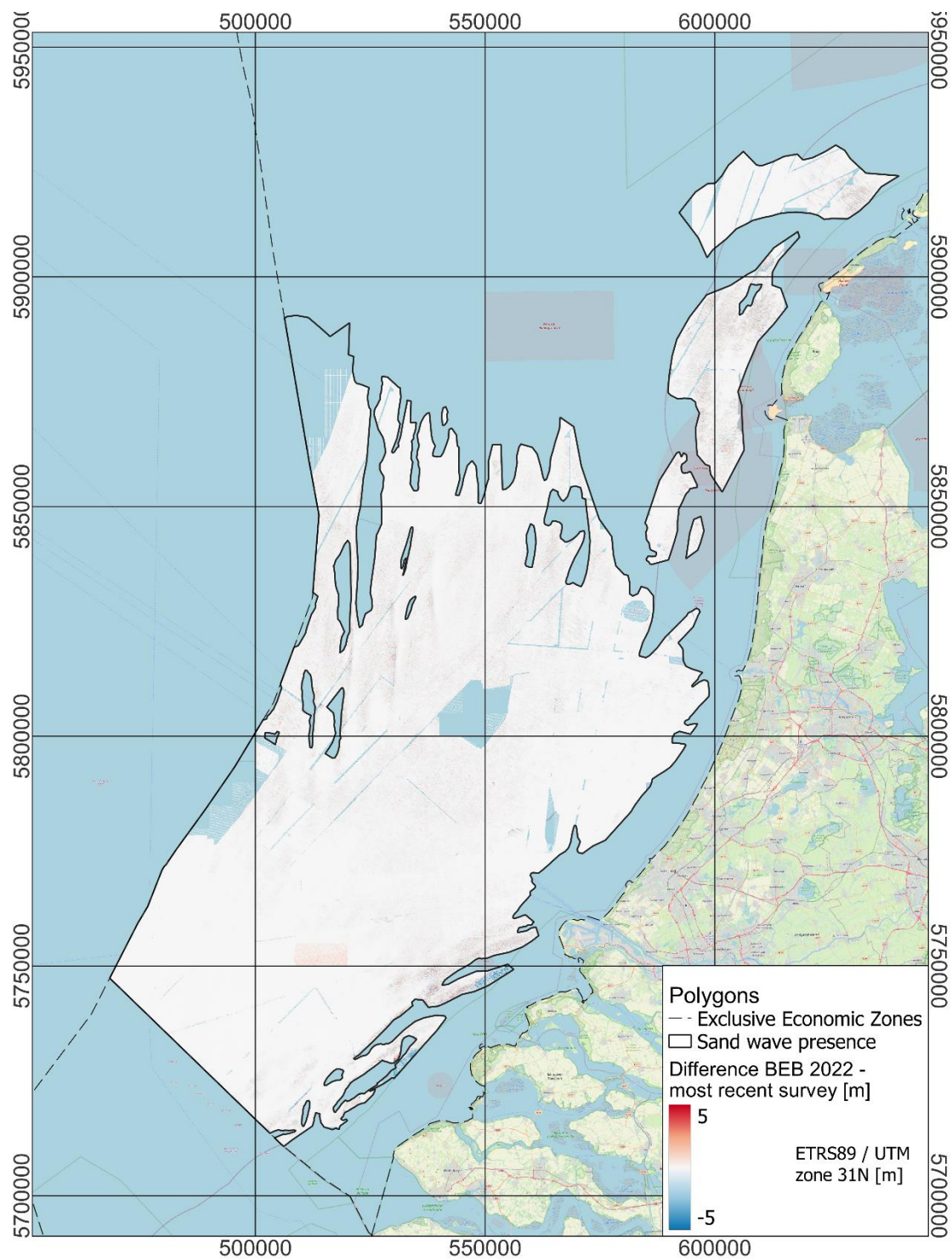


Figure 4.8: Sand wave area with the difference between the BEOL for 1945 and the most recent measured seabed.

4.4 Expected coverage of objects

Output of the hindcast as discussed in previous sections is translated to maps of expected coverage of objects. In this section the minimum, mean and maximum thickness of the sediment above an expected CE for the sand wave area in the DCS are shown in Figure 4.9 to Figure 4.11, respectively. The methodology to derive these maps is discussed in Section 3.7.

In most of the areas the larger burial depths are located around the crest locations of the sandwaves. For the mean estimation of the thickness above objects in 3 % of the area objects are expected at the surface, in 63 % the thickness is up to 1 m, in 20 % between 1 and 2 m, in 8 % between 2 and 3 m and in 4 % between 3 and 4 m.

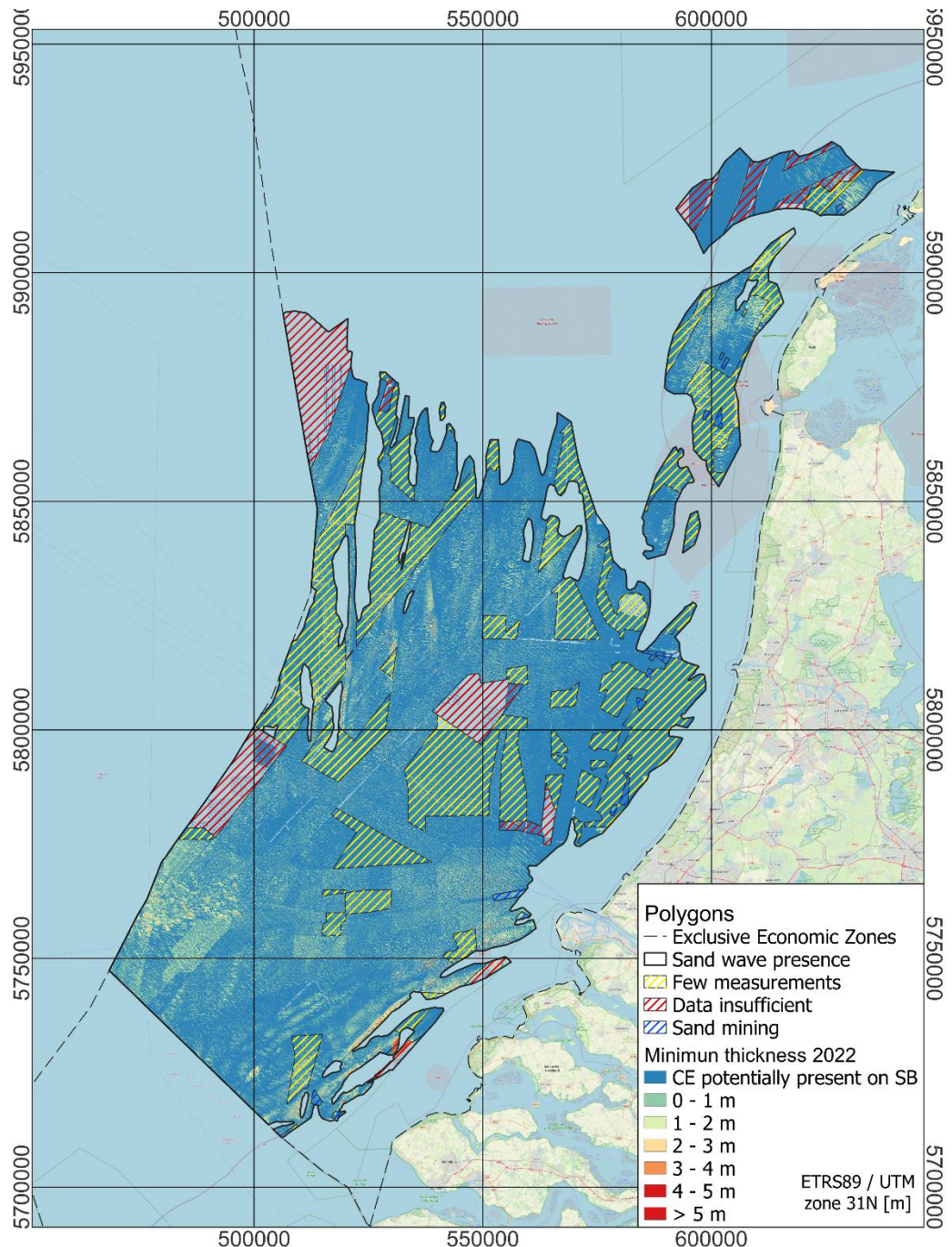


Figure 4.9: Sand wave area with the minimum thickness of the sediment above an expected CE. The minimum thickness is calculated by means of: $BEB2022 - \min(HOL1945; HSBL2022)$.

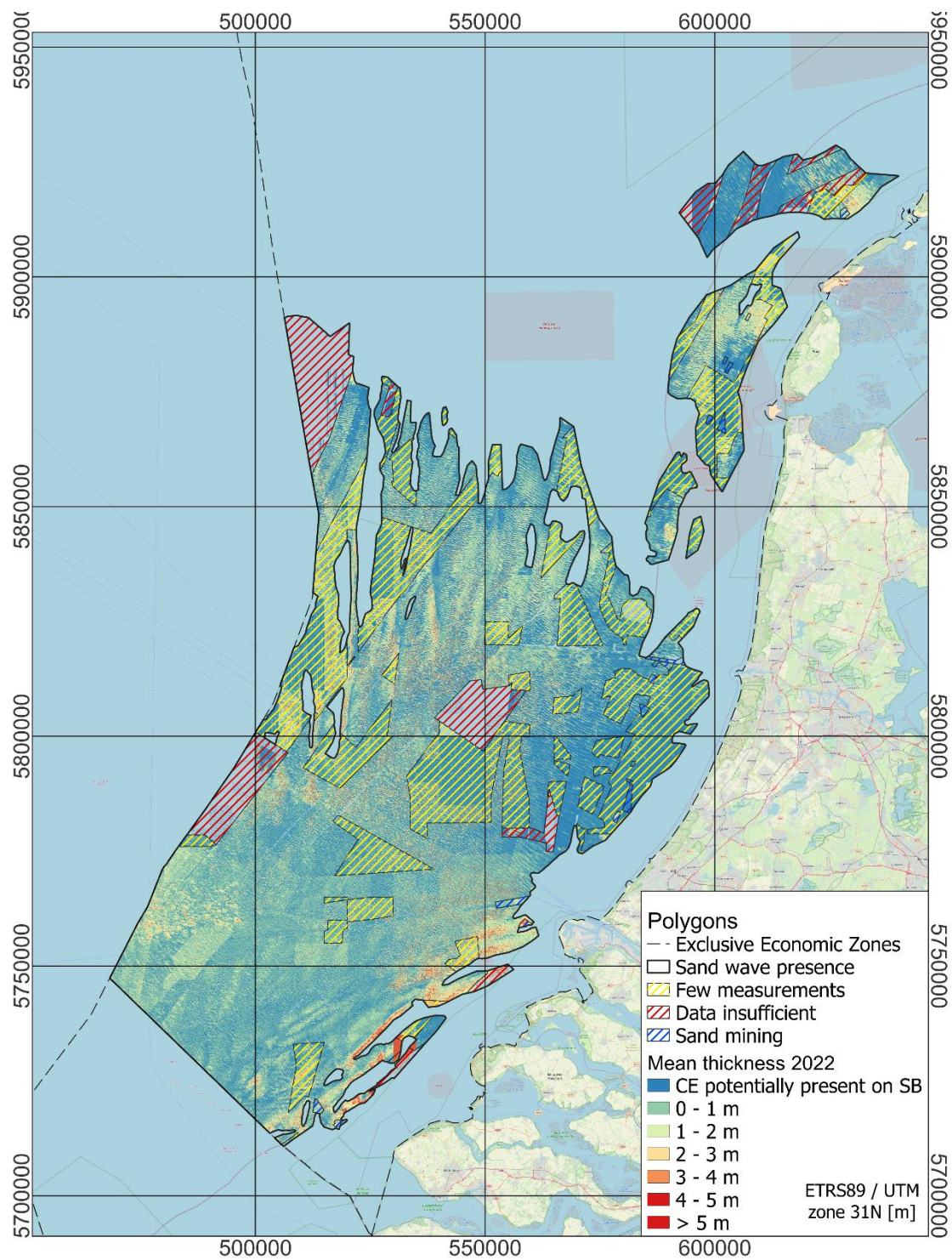


Figure 4.10: Sand wave area with the mean thickness of the sediment above an expected CE. The mean thickness is calculated by means of: $BEB_{2022} - \min(BEOL_{1945}; BSL_{2022})$.

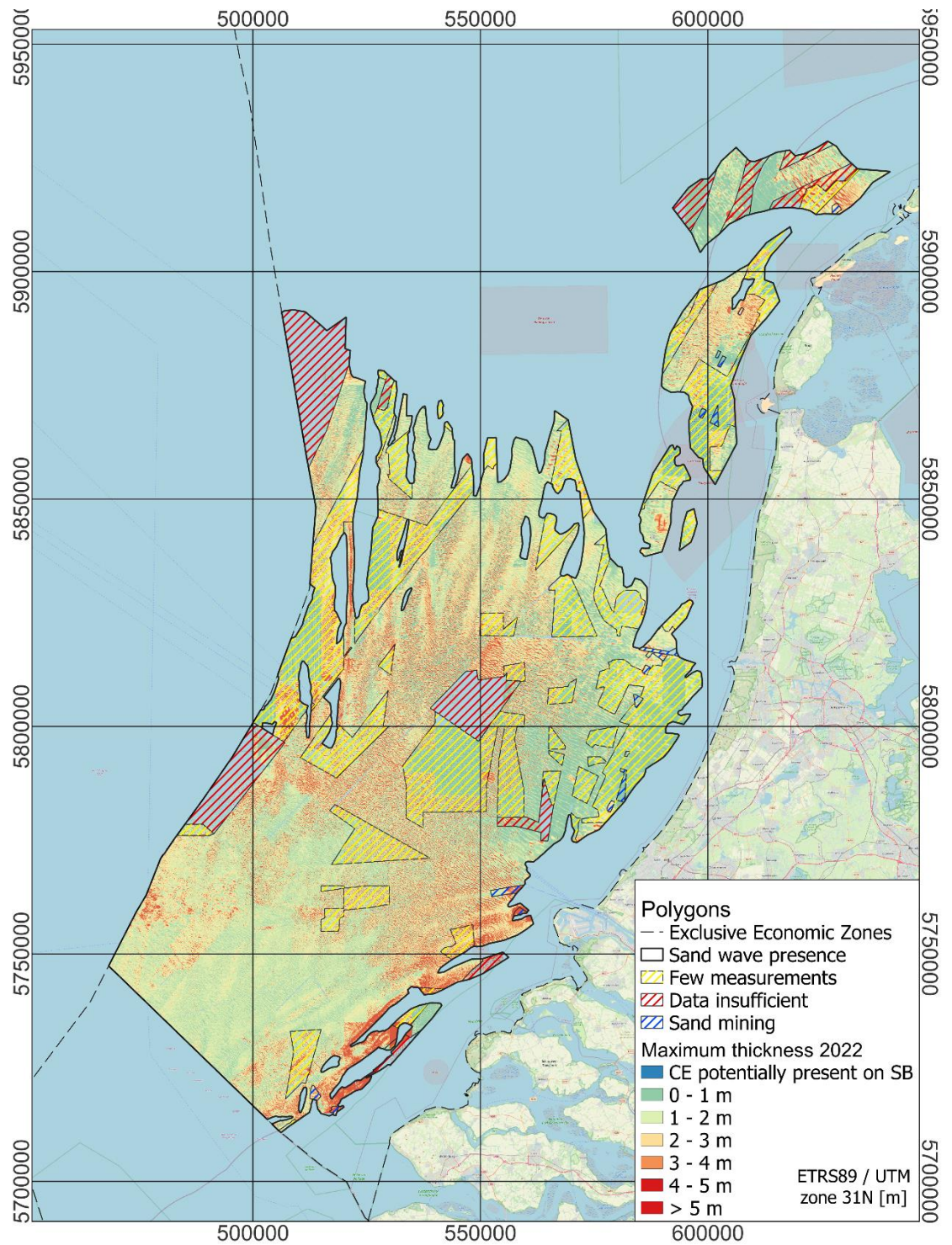


Figure 4.11: Sand wave area with the maximum thickness of the sediment above an expected CE. The maximum thickness is calculated by means of: $BEB2022 - \min(LOL1945; LSBL2022)$.

4.5 Quality map

The analysis provided insight in the dynamics of sand waves on the DCS. It was found that the quality of the results is subject to a number of limitations. For example, limited spatial coverage of survey data, quality of available data and human interventions can significantly influence results. In Section 3.8 insight is given to the quality checks performed on the results. These checks resulted in exclusion of certain datasets because of limited quality.

The quality of the final results is presented in Figure 4.12. The map highlights areas with limited amount of survey data available (red and yellow) and areas where human interventions such as sand mining have taken place (purple). Areas shaded green are areas for which at least three measurements are available, and no human interventions were observed. Quality of results in these areas is highest.

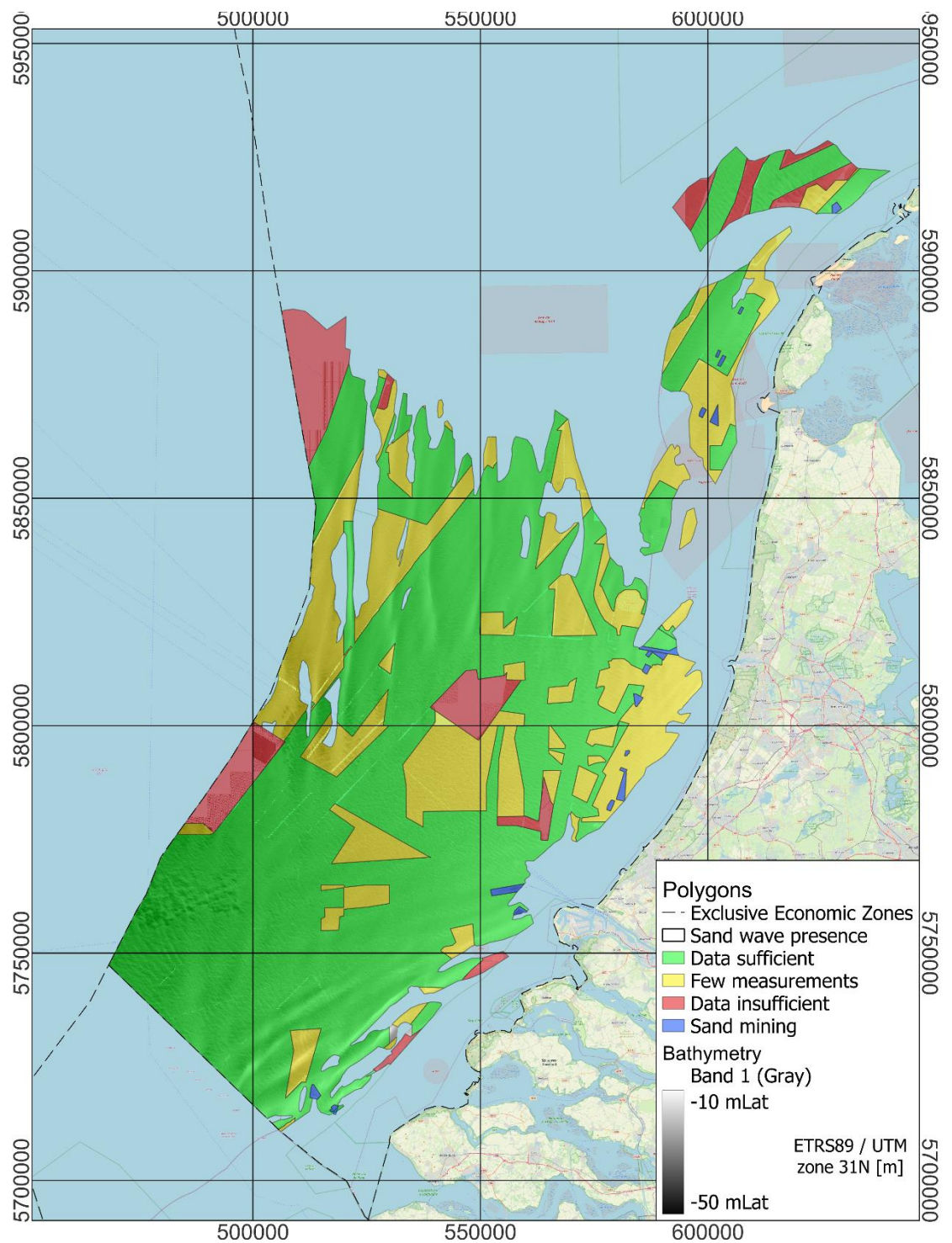


Figure 4.12: Quality map of the sand wave area showing the areas with sufficient data, few measurements, insufficient data and areas where sand mining has been performed in the past.

5 Discussion and recommendations

The presented assessment covers hindcasted and predicted historic and future seabed levels based on a data-driven analysis. Similar morphodynamic assessments often focus on a single area or wind farm zone and can therefore be detailed. Upscaling of the analysis to the entire DCS resulted in the need for a significant amount of quality checks. To discuss the results of the presented analysis, this chapter includes a discussion and recommendations.

5.1 Discussion

This morphodynamic assessment focusses on characterisation of seabed dynamics to quantify seabed level changes over the period 1945 to 2022 and over the period 2022 to 2050 for the DCS. Different levels are provided describing the bandwidth in Lowest Seabed Levels for given years. This study focusses solely on the areas where sand waves are present on the DCS.

The analysis conducted highlighted that the DCS is characterised by significant areas of sand waves particularly in the southern part of the DCS. Furthermore, the area is characterised by tidal sand banks in the north and south, with the northern banks running from north to south and the southern banks facing the coast of Zeeland. The sand waves typically migrate to the north-east, and locally to the south-west, with rates ranging from 0 to 3 m/year. In the south-western part of the sand wave area, migration rates are low. Meanwhile, the north-eastern region shows higher migration rates which gradually increase closer to the Wadden Islands. The highest migration rates are found near coastal areas and on top of sand banks, which significantly redirect the migration direction of the sand waves in several areas.

The analysis, but also detailed morphodynamic assessments on wind farms on the DCS, highlighted that spatial variability in seabed dynamics can be quite significant locally. This level of detail is to be considered in conjunction with the upscaling of the morphodynamic analysis to the entire DCS. Locally the spatial variability (spatial differences in sand wave migration rates and directions) might be captured in less detail compared to the detailed assessments.

The hindcast and predictions of future seabed levels is based on trends derived from the seabed mobility analysis. The resulting seabed levels are created based on a number of assumptions as described below. These assumptions need to be considered when using the hindcasted and predicted seabed levels.

It is assumed that relatively small objects, such as CE, cause no/negligible flow disturbance themselves and, therefore, will only cause local scour that can result in partial settlement of the object. These objects will, however, not affect the processes responsible for sand wave dynamics. Therefore, they will experience coverage if a sand wave crest passes over.

After the initial impact, a CE in a sandy environment might partially self-bury due to scouring (Menzel et al., 2022). It is also expected to experience burial and/or exposure due to bedform migration or large-scale seabed level changes. Under the effect of these morphodynamic processes the CE is assumed to stay at the lowest seabed level it has experienced between 1945 and now. Quantification of initial penetration of CE into the seabed is not part of the scope of this study. If significant penetration occurred during impact, then at locations that mainly experienced seabed rising the actual vertical level of the CE may be overestimated. Additionally, potential movement of an exposed CE under hydrodynamic or wave forcing is not assessed.

The BEB only provides a very rough indication of the possible seabed development during the predicted period and should not be treated as a firm design parameter. For design, the LSBL and HSBL provide better information (maximum expected potential seabed level variations at each grid point).

The seabed close to possible scour protections and cable crossings will most likely lower more than the LOL and LSBL levels because of local scour and edge scour. Buried electricity cables that cause no flow disturbance will only incur this scour effect if the cables are exposed on the seabed.

The presented seabed levels are a valuable dataset to (further) decrease the risks related to CE. Despite the limitations, as can be expected from an extrapolation of 60 to 75 years, the resulting surfaces and maps will help in assessing the likelihood of presence of CE at and under the seafloor.

The predicted seabed level changes presented in this study follow from the applied morphological analysis techniques, describing the (uncertainty of the) physics and the natural variability of the analysed morphological system. No additional safety margins for design purposes have been applied.

5.2 Recommendations

The presented hindcasted and future seabed levels are a result of the applied methodology. This methodology was previously successfully applied in detailed morphodynamic assessments for wind farm zones on the DCS. Because of local characteristics, upscaling of the analysis to the entire DCS resulted in increased uncertainties of historic and future seabed levels as compared to detailed morphodynamic assessments. The presented seabed levels provide a good insight in historic and future bandwidths of seabed levels, however detailed assessments are recommended in case depths of buried objects and cable burial depths need to be determined with more accuracy. These detailed studies can be useful for e.g. sand mining activities or assessment of a cable trajectory.

To keep the morphodynamic analysis up to date it is advised to periodically update the database following input of new bathymetric data. Especially areas which are now marked as of lower quality, either due to limited amount of data or low quality of available data, are recommended to be updated. During such an update, also improved analysis methods can be introduced with the aim to further quantify uncertainties. It is highly recommended that any additional bathymetry dataset is provided in a high resolution, at least 1.0 by 1.0 m, which resolve both sand waves and megaripples.

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